Locomotor system 15

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Introduction

Musculoskeletal symptoms are a major cause of pain and disability, accounting for a quarter of all general practitioner consultations in the United Kingdom (UK), with significant economic consequences. Common musculoskeletal conditions, such as back pain and osteoarthritis, are the dominant causes of chronic pain, disability and work loss in the UK and many other countries, consuming considerable health and social service resources. In addition, inflammatory arthritis and connective tissue disorders may present with musculoskeletal symptoms. As such, it is critical that clinicians are able to assess patients with such symptoms and initiate appropriate management or referral to a specialist centre. Early diagnosis and treatment of inflammatory arthritis, such as rheumatoid arthritis, has been consistently shown to improve patients' long-term outcome with the early initiation of disease-modifying therapy. The autoimmune rheumatic disorders, although much less common, cause significant morbidity with the potential for end-organ damage which may be fatal if not recognized and treated early.

The objectives of performing a musculoskeletal assessment are to:

- Make an accurate diagnosis
- Assess the severity and consequences of the condition
- Construct a clear management plan

Taking an effective structured history and making a simple, focused examination are likely to be more important than imaging and serology, which on their own may be falsely reassuring. Although modern musculoskeletal medicine uses complex imaging and immunological investigations, in most patients with locomotor disorders, diagnosis can be achieved at the bedside without complex investigations.

General assessment: the 'GALS' locomotor screen

This brief screening examination, which should take 1 to 2 minutes, has been devised for use in routine clinical assessment. This brief global assessment of the gait, arms, legs and spine (GALS) is typically used by both non-specialist and specialist alike (Box 15.1). The routine requires some practise to become proficient, but once achieved, it has been shown to be highly sensitive in detecting significant abnormalities of the musculoskeletal system. It involves inspecting carefully for joint swelling and abnormal posture, as well as assessing the joints for normal movement. Typically, the clinician will then focus on particular areas of interest with more dedicated and targeted examinations.

Screening history

If answers to the following four questions are negative, a musculoskeletal disorder is unlikely:

1. Do you have any pain or stiffness in your muscles, joints or back?
2. Have you ever had gout or arthritis?
3. Can you dress yourself completely without difficulty?
4. Can you walk up and down stairs without difficulty?

Positive answers imply the need for a more detailed assessment, as will be described in this chapter.

Screening examination

Gait

Watch the patient walking and turning back towards you. Normal: symmetry, smooth movement, arm swing, no pelvic tilt, normal stride length, ability to turn quickly.

With the patient standing in the anatomical position, observe from behind, from the side and from in front. Muscle symmetry should be noted along with limb alignment. Spine and limb alignment can be observed as well as abnormalities in the feet, such as pes planus or pes cavus.

Common gait abnormalities are listed in Table 15.1.

Spine

Inspect the standing patient. Normal on inspection: no scoliosis, symmetrical paraspinals, normal shoulder and gluteal muscle bulk and symmetry,

Table 15.1 Gait abnormalities

Type	Description	Causes
Antalgic	Avoiding weight bearing on affected side	Pain related to arthritis or tendonitis
Spastic	Leg swings outwardly from hip with extended knee	Hemiplegia related to stroke
Shuffling	Small shuffling steps, speeding up (festination), reduced arm swing and difficulty turning	Parkinson's disease
Ataxic	Wide-based gait, swaying torso	Cerebellar disease. Peripheral sensory ataxia (typically worsens with loss of visual input)
Waddling (Trendelenburg)	Exaggerated lateral movement of trunk and circumduction of hip to compensate for weak abductors	Bilateral gluteus medius tendinopathy
High stepping (foot drop)	Excessive hip and knee flexion to accommodate failure of ankle dorsiflexion	Peroneal neuropathy (multiple causes)

Box 15.1 GALS checklist

Gait

- Observe gait
- Observe patient in anatomical position

Arms

- Observe movement: hands behind head
- Observe backs of hands and wrists
- Observe palms
- Assess power grip and strength
- Assess fine precision pinch
- Squeeze metacarpophalangeal joints (MCPJs)

Legs

- Assess full flexion and extension
- Assess internal rotation of hips
- Perform patellar tap
- Inspect feet
- Squeeze metatarsophalangeal joints (MTPJs)

Spine

- Inspect spine
- Assess lateral flexion of neck
- Assess lumbar spine movement

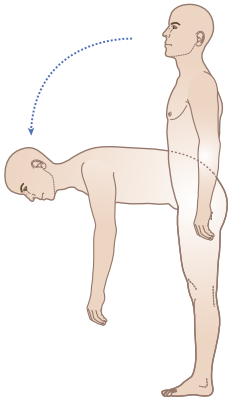


Figure 15.1 Inspect patient from behind and side, observing for normal spinal curves, then ask the patient: 'bend forwards to try to touch your toes'.

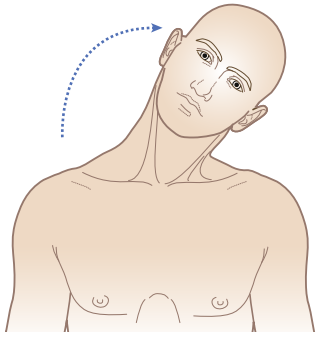


Figure 15.2 From the front, ask the patient: 'place your ear on your right then your left shoulder'.

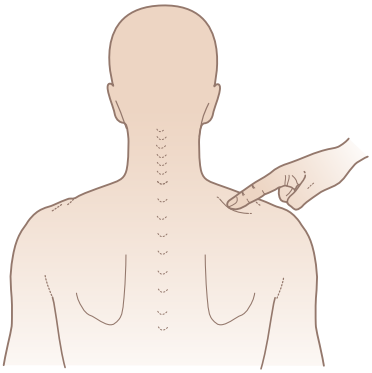


Figure 15.3 Gently press the mid-point of each supraspinatus to elicit the hyperalgesia of fibromyalgia.

level iliac crests, normal cervical and lumbar lordosis. Movement: finger to floor distance less than 15 cm, lumbar expansion greater than 6 cm, ear touches acromion.

- From behind: look for abnormal spinal and paraspinal anatomy and look at the legs.
- From the side: look for abnormal spinal posture, then ask the patient to bend down and touch his toes (Fig. 15.1).

- From the front: ask the patient to 'put your ear on your left then right shoulder' and watch the neck movements (Fig. 15.2).
- Gently press the mid-point of each supraspinatus muscle to elicit tenderness (Fig. 15.3).

Arms

Ask the patient to follow instructions as in Table 15.2 (see Figs 15.4–15.9).

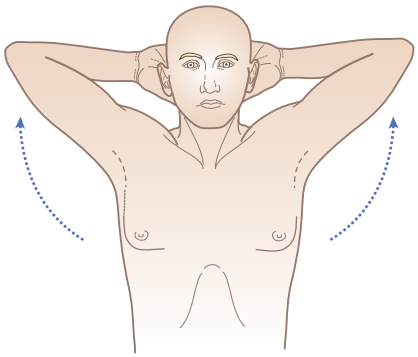


Figure 15.4 Ask the patient: 'Put your hands behind your head, elbows back'. Observe for pain or restricted movement.

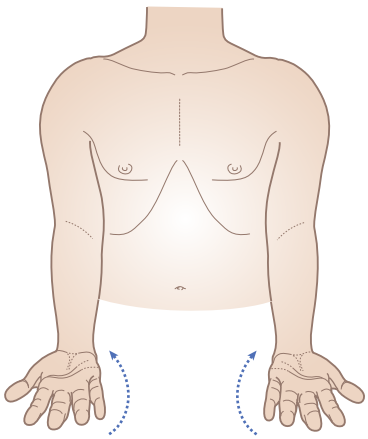


Figure 15.5 Ask the patient: 'Put your hands out, palms down, and then turn your hands over'.

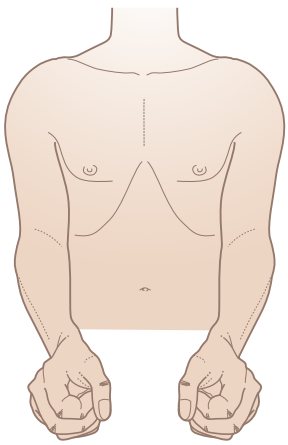


Figure 15.6 Ask the patient: 'Make a fist with both hands'.

Table 15.2 Instructions for examining arms

Instruction	Normal outcome
'Raise arms out sideways and up above your head'.	180° elevation through abduction without wincing
'Touch the middle of your back'.	Touches above T10 with both hands
'Straighten your elbows right out'	Elbows extend to 180° or slightly beyond (females) symmetrically
'Place hands together as if to pray, with elbows right out'.	90° wrist extension and straight fingers
'The same with hands back to back'.	90° wrist flexion
'Place both hands out in front, palms down, fingers out straight'.	No wrist/finger swelling/ deformity, 90° pronation
Metacarpophalangeal (MCP) cross-compression	No tenderness
'Turn hands over'	90° supination No palmar swellings, wasting or erythema
'Make a fist and hide your nails'.	Can hide fingernails
'Pinch index, middle finger and thumb together'	Can do
Also look for swelling between the heads of the metacarpals and gently squeeze across the MCP joints to elicit tenderness.	Tenderness may be elicited

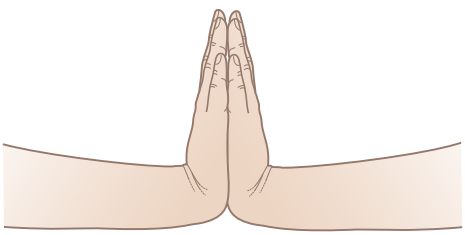


Figure 15.7 Ask the patient: 'Place your hands together as if to pray, with elbows right out'.

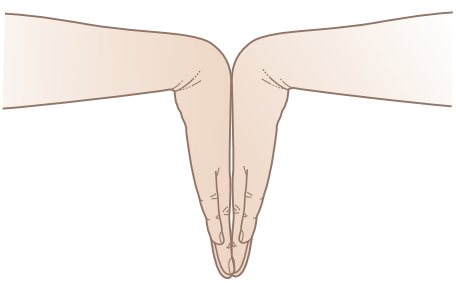


Figure 15.8 Ask the patient: 'Do the same with your hands back to back'.



Figure 15.9 Look for swelling between the heads of the metacarpals and gently squeeze across the metacarpophalangeal joints to elicit tenderness.

Legs

With the patient still standing:

- Examine the lower limbs for swelling, deformities or limb shortening. Normal: no knee deformity, anterior or popliteal swelling, no muscle wasting, no hindfoot swelling or deformity.
- Then, with the patient lying on a couch, continue the examination following the instructions in Table 15.3 (see Figs 15.10–15.14).

Table 15.3 Instructions for examining legs

Instruction	Normal outcome
Flex hip and knee, holding knee	No bony crepitus, 140° knee flexion
Passively rotate hips	90° total pain-free rotation
Bulge test/patellar tap	No detectable fluid
Palpate popliteal fossa	No swelling
Inspect feet	No deformity, callosities or forefoot widening (daylight sign)
Test subtalar and ankle movement	Pain-free calcaneal mobility at subtalar joint, dorsiflexion beyond plantigrade and 30° plantarflexion



Figure 15.10 Gently flex the hips and knees, feeling for crepitus at the knee during movement, and look for pain and restriction of movement. Look for knee effusion.

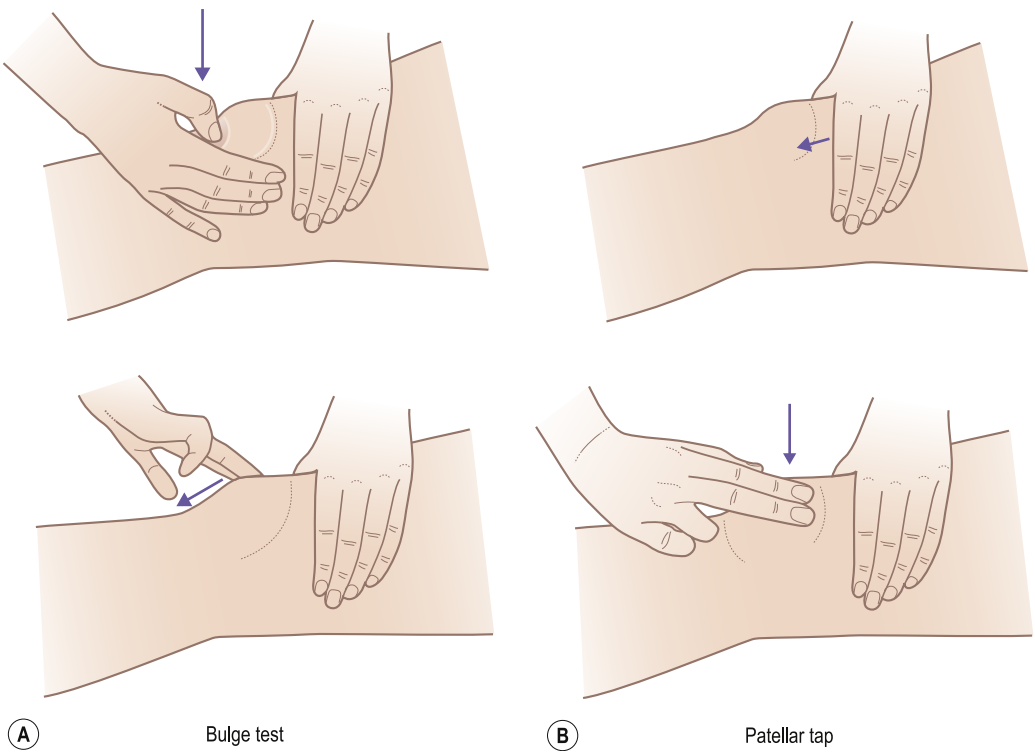


Figure 15.11 **(A)** Bulge test and **(B)** patellar tap.

This examination can be conducted in approximately 2 minutes, especially if the clinician performs the movements and asks the patient to follow them. The precise order of the examination is not important, and clinicians usually develop their own pattern of examination.

Children

Children should have their height and weight measured and plotted on centile charts to assess growth. When examining children, the assessment of the musculoskeletal system should be done with a parent present and include all the components of the adult version, with minor additions as follows:

Ask the child to walk on his tiptoes and then on the heels.

From the front, ask the child to put his hands together (as if praying) and also put his hands back to back and then have him reach his arms up towards the sky.

Additionally, when assessing the spine, ask the child to open his mouth and insert three of his own fingers into it.

Symmetry is key and any asymmetry should be assessed for pathology. It is normal for toddlers to be 'bow-legged,' 'knock-kneed' and 'flat-footed'. Various foot and toe appearances are also normal, including in-toeing, out-toeing and 'curly' toes. These appearances usually resolve as the child grows.

Recording the results

The incorporation of a brief musculoskeletal examination into the routine examination should lead to a more detailed assessment if any abnormalities are found (Table 15.4).

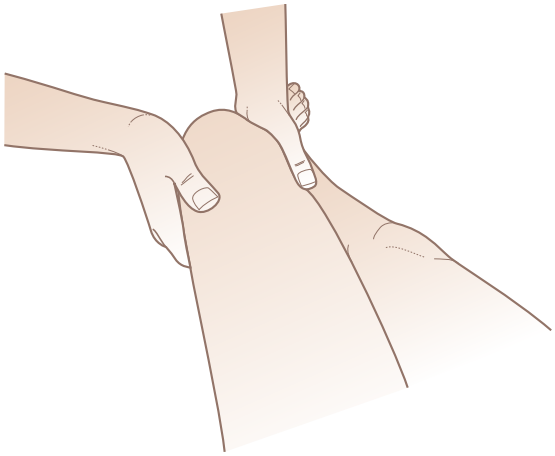


Figure 15.12 Palpate popliteal fossa.

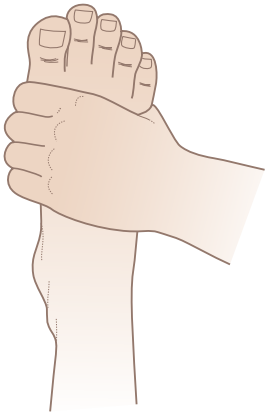


Figure 15.14 Squeeze across metacarpophalangeal (MTP) joints and inspect the soles of the feet.

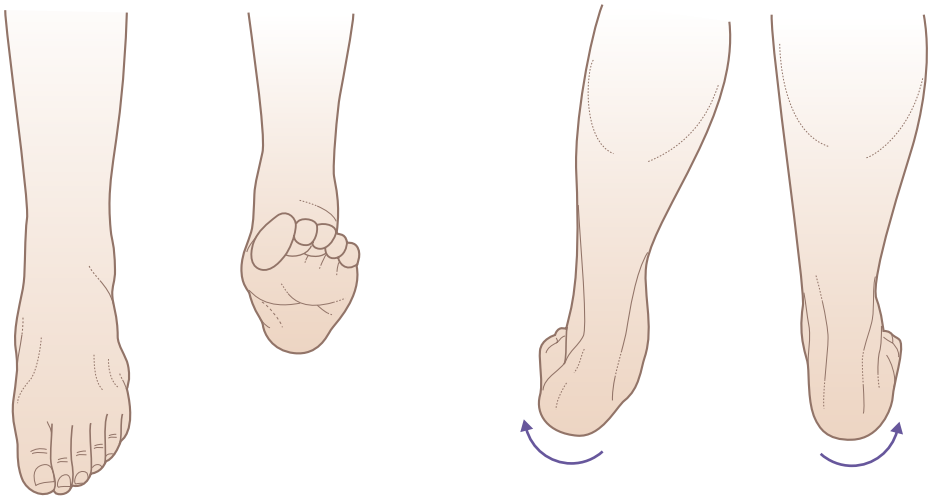


Figure 15.13 Test subtalar and ankle movement.

Table 15.4 How to record the locomotor screen routinely in the medical record

Screening issues:		Pain	0
		Gout or arthritis	0
		Dressing	✓
		Walking	✓
with positive answers being elaborated.			
Examination:			
A (Appearance)		M (Movement)	
G	✓		
A	✓	✓	
L	✓	✓	
S	✓	✓	
with abnormal findings indicated, e.g. in rheumatoid arthritis:			
A (Appearance)		M (Movement)	
G	✗		
A	✗	✗	
L	✗	✗	
S	✓	✓	
Slow painful gait.			
Synovitis of metacarpophalangeal (MCP), metatarsophalangeal (MTP), wrists and knee joints.			

Specific locomotor history

General demographic details, such as age, sex and occupation, should be recorded. Start the full musculoskeletal history with the presenting complaint and ask the patient to describe the sequence of symptoms, related features and events since the onset. Characteristic diagnostic points may emerge. Pain is a common presenting complaint and, as such, the clinician is expected to explore this fully. Using the WILDA (*W*ords, *I*ntensity, *L*ocation, *D*uration, *A*ggravating/alleviating factors) pain assessment approach may be helpful in drawing out key aspects of the patient's history and clarifying the nature of the complaint (see also Chapter 11).

Words

'I have pain' is not descriptive enough to inform a clinician about the character of the pain symptoms. Patients may have more than one type of pain and should be encouraged to describe their pain as much as possible. Neuropathic pain can be described as burning, shooting, tingling, radiating, lancinating or numbness. Somatic pain may be described as dull or throbbing and can usually be well localized. Somatic pain is most commonly described in the context of musculoskeletal pain afflicting joints and the spine.

Pain described as squeezing, pressure, cramping, distention, dull, deep and stretching is visceral in origin.

Box 15.2 Common patterns of joint pain radiation

Pain may be referred or radiate from an affected joint, e.g.:

- Neck pain may radiate through the occiput to the vertex or to the shoulder and down the arm, with paraesthesiae if there is nerve-root impingement.
- Shoulder joint pain may radiate to the elbow and below.
- Thoracic spine nerve root pain may radiate around the chest and mimic cardiac pain.
- Lumbar spinal root pain may radiate through the buttock and leg to the knee and below, with paraesthesiae in the foot with a large disc prolapse; this is called 'sciatica'.
- Hip-joint pain may radiate to the knee and below.
- Knee pain may radiate above and below the joint.

Intensity

It may be useful to use a simple visual analogue scale or ask the patient to rate his pain on a 0 to 10 scale. This may allow for monitoring and assessing progression of the underlying disorder.

Location

An accurate location of pain is often very helpful, although it must be remembered that joint pain may radiate. Patients should be asked, 'Where is your pain?' or 'Do you have pain in more than one area?'

Pain may be referred or radiate from an affected joint (Box 15.2).

Duration

The temporal description of pain is important in understanding the root cause. Pain may be constant, episodic or a combination of both. Is the patient ever pain free? Can he clearly identify the time of onset and is there a pattern of periodicity? For example, an osteoporotic vertebral fracture may be acute and self-limiting, which contrasts with the severe, constant and gnawing pain of a spinal metastatic deposit that prevents sleep.

Aggravating/alleviating factors

Aggravating and alleviating factors are often very helpful when trying to make a diagnosis. Back pain or stiffness that improves with exercise may point to an inflammatory cause, whereas pain that worsens after use is often described in the setting of degenerative arthritis.

Back pain is the most common musculoskeletal complaint; as such, it deserves further consideration. When taking a history of a patient's back symptoms

Table 15.5 Red flag symptoms associated with back pain with potential associated causes

Red flag	Potential cause
Elderly (>65 years) or young (<40 years) patient	Elderly patient: osteoporosis Young patient: inflammatory arthritis
Immunocompromised	Infection
Trauma	Fracture or displacement
History of cancer	Malignancy/metastasis
Nocturnal pain	Malignancy
Systemically unwell (temperature/weight loss/night sweats)	Infection
Thoracic pain	Inflammatory: ankylosing spondylitis Infection/malignancy Aortic dissection/ pancreatitis
Saddle anaesthesia/ bilateral sciatica/progressive neurological deficit	Cauda equina or cord compression
Bladder, bowel or sexual dysfunction	Cauda equina or cord compression

it is important to identify potential 'red flags,' which mandate further imaging or investigations, to exclude a potentially sinister underlying diagnosis. If red flag signs are present, then referral to a specialist for further evaluation is advised and a patient should be advised to rest and to avoid physical activity until then. Importantly, nerve root pain (unilateral sciatica) is not itself a cause for alarm, and conservative treatment should be effective. A list of red flag symptoms associated with back pain can be found in Table 15.5, with potential associated causes.

Pain may be referred or radiate from an affected joint (see Box 15.2). With these pain patterns in mind, the joints above and below the apparently affected area should be examined.

Chronic pain syndromes

Chronic pain syndromes need to be distinguished from other rheumatic disorders and a careful history is essential. A number of pain descriptors used by patients can indicate the influence of non-organic amplifying factors. For example, at least in the English language, words like 'searing,' 'torturing' and 'terrifying' can be used by patients with chronic widespread pain and fibromyalgia. Fibromyalgia falls under an umbrella term—Central Sensitization Syndromes (CSS) and so may present with overlapping symptoms with other conditions such as irritable bowel syndrome, chronic fatigue and temporo-mandibular joint (TMJ) disorders. These debilitating disorders are often accompanied by poor sleep patterns, severe fatigue and depressed mood. The difficulty of cultural, religious and linguistic influences in interpreting pain should not be underestimated.

Joint disease

A combination of pain and stiffness, leading to loss of function, is a classic feature of joint disease. Usually one component predominates, as with stiffness in inflammation and pain in mechanical joint problems. Therefore, specific questions will establish whether symptoms are non-inflammatory (e.g. osteoarthritis) or inflammatory (e.g. rheumatoid arthritis). As with any condition, the good clinician will also draw out the consequences of the disorder in terms of the impact on the patient's day-to-day function and quality of life. The effect on relationships, work and social activities will help the clinician to gauge the severity of the problem and possibly elicit issues of stress and psychosocial pressure.

Non-inflammatory joint disease

Pain of a non-inflammatory origin is more directly related to function and use and usually is relieved with rest. This is in contrast to inflammatory joint pain, which is often present at rest as well as in use, and tends to vary from day to day and from week to week. Stiffness, particularly in the morning or after a period of inactivity, is often a feature in the history of inflammatory joint pain and typically lasts no more than 30 minutes. However, in severe cases the duration may be much longer. Subjective joint swelling may be described and indeed patients with knee osteoarthritis may present with a significant effusion. Joint swelling in this context tends to be more intermittent than the persistent joint swelling and warmth seen in inflammatory conditions.

Locking of a joint may occur. In the knee, this means that the knee becomes locked in such a way that it will not extend fully, although it may flex. In other joints, locking is less well defined and simply means that at some point through its range of motion the joint becomes stuck, usually associated with pain and often followed by swelling. Locking is caused by material within the joint interfering with movement at the articular surfaces. In the knee, this is usually part of one of the menisci or a cartilaginous loose body.

Inflammatory joint disease

Early morning stiffness

Early morning joint stiffness that persists for more than 30 minutes is an important symptom of active inflammatory joint disease. Ask about redness (rubor),

Box 15.3 Importance of distribution of joint involvement in differential diagnosis of pauciarthritis or polyarthritis

Typically symmetrical

- Upper and lower limbs: rheumatoid arthritis, systemic lupus erythematosus
- Especially upper limbs: haemochromatosis (hand: index, middle metacarpophalangeal joints)

Typically asymmetrical

- Psoriatic arthritis

Typically lower limb and asymmetrical

- Spondyloarthritis (e.g. ankylosing spondylitis, reactive arthritis)
- Document any associated features

Box 15.4 Acute gout

A 48-year-old publican presented with sudden severe pain and swelling of his left big toe. He was unable to bear weight or wear a shoe and said that he could not even bear the weight of his bedclothes on the toe at night. His past history included hypertension, for which he was taking a thiazide diuretic, and his alcohol intake was excessive. On examination, he had tophi on the ears (see Fig. 15.15) and the left first metatarsophalangeal joint was red, hot, swollen and exquisitely painful to touch. Aspiration of the joint revealed urate crystals.



Figure 15.15 Gouty tophus on the ear. Other sites include the elbows, fingers and toes.

warmth (calor), tenderness/pain (dolor) and swelling (tumour), the classic features of inflammation.

Distribution of joint disease

The pattern of joint involvement is important. Common patterns are monoarticular (single joint), pauciarticular (up to four joints), polyarticular (many joints) and axial (spinal involvement). The pattern of joint involvement often aids in making a list of likely differential diagnoses. For example, rheumatoid arthritis is typically a symmetrical inflammatory arthritis of the small and large joints, often presenting with metacarpophalangeal (MCP), wrist and metatarsophalangeal (MTP) joint involvement. There are only a few causes of an exactly symmetrical arthropathy (Box 15.3). An asymmetrical inflammatory arthritis presentation is often seen in spondyloarthropathies, such as reactive arthritis. An inflammatory monoarthritis presentation must arouse a suspicion of infection alongside other potential conditions, such as gout or psoriatic arthritis. A history of fever or sweating may be helpful and, if possible, synovial fluid should be aspirated from the joint to look for an infection. Monoarthritis will certainly raise the suspicion of an infected joint, but may also be seen in the context of crystal arthropathies and spondyloarthropathies, particularly psoriatic arthritis in which the knee or elbow is often singularly affected. Other conditions have such a classic history that it is almost diagnostic. For example, acute inflammation in the first metatarsophalangeal joint (hallux) suggests a diagnosis of gout (Box 15.4). The pain and stiffness of the shoulder and hip girdles in polymyalgia rheumatica is also typical (Box 15.5). The pattern of the spondyloarthropathies can also be diagnostic with inflammatory sacroiliac and spinal pain and stiffness, lower limb arthritis, Achilles tendinitis and plantar fasciitis.

Box 15.5 Polymyalgia rheumatica

A 74-year-old woman presented with a 2-month history of pain around her hips and shoulders. The key points in the history were that these regions were extremely stiff for 4 hours each morning and an inability to turn over in bed, and that there was no joint swelling. Clinically there was pain and some limitation of movement at the shoulders and hips, but no synovitis. Her erythrocyte sedimentation rate was 96 mm/1st hour, and she had a dramatic response within 2 days to a moderate dose of prednisolone.

Box 15.6 Pointers in the history towards the differential diagnosis of monoarticular inflammatory joint disease

First attack

- Exclude infection by aspiration for culture and crystals
- If negative culture, but high-risk group, biopsy, e.g. tuberculosis in Asian immigrants or immunosuppressed patients

Recurrent attacks

- Flitting (gonococcal arthritis, rheumatic fever)
- Episodic (crystal arthritis, palindromic rheumatism)

Persistent synovitis with none of the above features

- Look for systemic features and check serology (e.g. rheumatoid factor or antinuclear antibody)

Recurrent attacks of joint pain

Ask if the same joint is always involved. If not, define the patterns of involvement, the severity and



Figure 15.16 Extensive rash in sun-exposed areas in a woman with systemic lupus erythematosus and anti-Ro antibodies.

duration of the episodes and any associated clinical symptoms. Crystal arthropathies tend to affect the same joints recurrently with the first MTP joint most common followed by ankle and knee.

Episodic joint pain

Ask if attacks of joint pain are acute and associated with redness around the joint, with the attacks lasting about 48 hours (occasionally up to 1 week) and migrating to other joints. This is typical of palindromic rheumatism, which may progress to rheumatoid arthritis. Recurrent painful swelling of the same joint may be indicative of a crystal arthropathy, such as gout or calcium pyrophosphate disease.

Flitting or migratory joint pains

The term 'flitting' or migratory joint pains is used to describe inflammation beginning in one joint and then involving others, usually one at a time for about 3 days each. Gonococcal arthritis should be considered; this is characterized by typical fleeting skin lesions and urethritis, in addition to joint pain. In rheumatic fever, there is associated cardiac involvement, and erythema marginatum and subcutaneous nodules may occur.

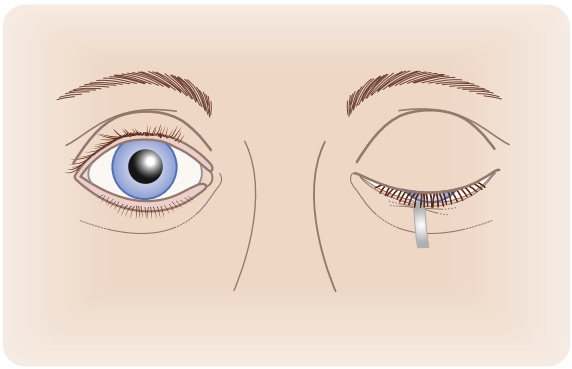


Figure 15.17 Schirmer's test. A sterile strip of filter paper is hooked over the lower eyelid. Less than 5 mm of wetness after 5 minutes is abnormal and is associated with autoimmune rheumatic disease.

The other features in the history that should be brought out are best considered under the differential diagnosis of polyarthritis (Box 15.6). Many arthropathies may have a monoarticular presentation.

Inflammatory connective tissue diseases

The autoimmune rheumatic disorders, such as systemic lupus erythematosus (SLE), Sjögren's syndrome, inflammatory myopathies, systemic sclerosis and the vasculitides, are multisystem disorders. It is critical that a full systems enquiry and physical examination are performed when inflammatory connective tissue is suspected. Associated features of these conditions include systemic symptoms, such as weight loss, malaise or fevers and rash, especially if the latter is photosensitive or vasculitic (Fig. 15.16). Alopecia, oral and genital ulceration, Raynaud's phenomenon and symptoms of neurological, cardiac, pulmonary and gastrointestinal involvement may occur. Dry eyes and dry mouth (sicca symptoms) are common and can be documented with Schirmer's test (Fig. 15.17). Renal disease is a serious complication of the autoimmune rheumatic disorders, especially in SLE and the systemic vasculitides. Clinical assessment should always include measuring blood pressure and dip-testing the urine for blood and protein and microscopy of the urine sediment for casts or dysmorphic red cells if this is positive. Ear, nose and throat involvement with sinusitis, facial pain and deafness are common in Wegener's granulomatosis and Churg–Strauss syndrome. A history of arterial and venous thromboses or miscarriages, especially in the context of livedo reticularis, should raise the suspicion of the antiphospholipid syndrome.

Soft tissue symptoms

Soft tissue problems, which are common, usually consist of pain, a dull ache, tenderness or swelling. In the elderly these symptoms often appear spontaneously, but in younger people there is

Box 15.7 Localization of the site of articular and extra-articular features

Joint
- Diffuse pain and tenderness
- Generalized joint swelling
- Restriction of movement, usually in all directions of movement (specific to each joint)

Tendon
- Localized pain/tenderness at attachment (enthesis) or in the tendon substance
- Swelling, tendon sheath or paratendon
- Pain on resisted movement
- Sometimes pain on stretch (e.g. Achilles)

Ligament
- Localized pain/tenderness at attachment or in ligament substance
- Pain on stretch
- Instability, if major tear

Bursa
- Localized tenderness
- Pain on stretching adjacent structures

Muscle
- Localized or diffuse pain and tenderness
- Pain on resisted action
- Pain on stretch (e.g. hamstring)

Box 15.8 Examination of the musculoskeletal system

General observations
- Gait
- Posture
- Mobility
- Deformity
- Independence: use of wheelchair or walking aids
- Muscle wasting
- Long bones

Fractures
- Joints
- Tendons
- Skin

usually a history of injury or overuse, through either occupation (e.g. tenosynovitis of the long flexor tendons of the hand) or sport (e.g. Achilles tendinitis in runners, shoulder rotator cuff tendinopathy in swimmers or throwers). It is important to define the exact site of the symptoms and the factors that either make them worse or induce relief. The mechanism of injury is also important and can help guide a focused examination. The localization of symptoms to specific soft tissue structures can be confirmed by careful examination (Box 15.7). The possible structures involved are joint, tendon, ligament, bursa and muscle. Local palpation may be enough to locate



Figure 15.18 Thenar wasting owing to carpal tunnel syndrome. This is often associated with osteoarthritis. Note nodal change on the terminal interphalangeal joints of the index fingers.

bursitis or enthesitis (inflammation at the attachment of a ligament or tendon to bone) accurately. A careful examination may reveal that the complaint and symptoms do not arise from joint pathology, but from soft tissue tendon or ligament structure next to the joint. This often changes the differential diagnosis. Movement of the joint region often elicits pain or discomfort in a particular direction, which can direct the clinician to the affected structure. Subacromial bursitis typically produces pain when the shoulder is abducted to 80° and resolves at 130° to 140°, described as a painful arc, whereas anterior and posterior movements are maintained.

The bones

Bone pain is characteristically deep seated and localized, but referred pain may confuse the clinical picture. In the case of fractures, unless pathological, almost always there will be a history of injury. In athletes, however, a fracture may be owing to chronic overuse, as in stress fractures of the tibia or metatarsals in runners. The spontaneous onset of pain may suggest Paget's disease (with bony enlargement, e.g. skull or tibia) or metastatic deposits. Infection must also be considered, particularly in younger patients or in immunodeficiency states. Consider also congenital or familial disorders as predisposing factors, for example multiple osteochondromata or brittle bone disease (osteogenesis imperfecta).

Examination

General Principles

Observe the patient entering the room (Box 15.8). Abnormalities of gait and posture may provide clues that can be pursued in history taking. Observation of any difficulty in undressing and getting onto the

examination couch will further help in assessment. The patient must always be asked to stand and walk, even when it is obvious that this may be difficult. Note how much help the patient requires from others or from sticks, crutches, etc. The musculoskeletal system includes the muscles, bones, joints and soft-tissue structures, such as tendons and ligaments. Remember that although muscle wasting may be caused by primary muscle disease (e.g. polymyositis), it is more commonly secondary to disuse, perhaps because of a painful joint, or to neuropathy owing to nerve root compression or peripheral neuropathy (Fig. 15.18). Examination of the muscles is discussed further in Chapter 16.



Figure 15.19 Paget's disease of the right tibia. Note tibial bowing and bony enlargement **(A)** and bony enlargement sclerosis with some patchy porosis in the X-ray of the upper tibia **(B)**.

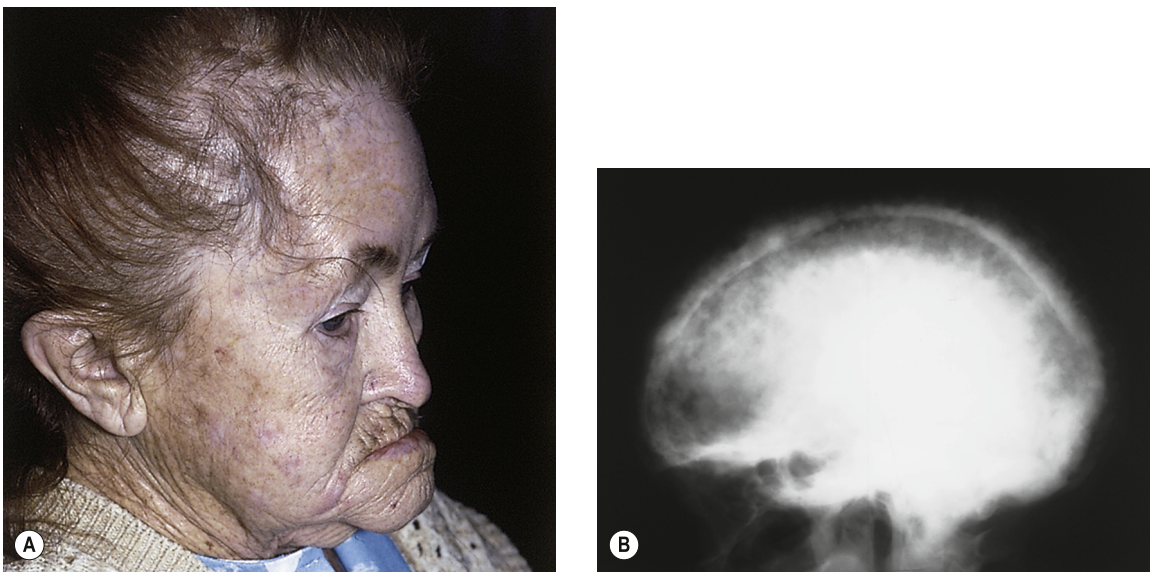


Figure 15.20 Paget's disease, causing deformity of the skull **(A)**. Note the thickened skull vault with remodelled bone **(B)**.

The bones

The examination of the bones should always be directed by information obtained from the history.

Inspection

Look for any alterations in shape or outline and measure any shortening. In Paget's disease (osteitis deformans), bowing of the long bones, particularly the tibia (Fig. 15.19) and femur, is associated with bony enlargement and, usually, increased local temperature. The skull is also commonly involved, but this may not be apparent until the disease is advanced. Early involvement of the skull bones can be detected on X-rays (Fig. 15.20). Alterations in the shape of bones also occur in rickets as a result of epiphyseal enlargement. Deformity of the chest in rickets is caused by osteochondral enlargement (rickety rosary).

Localized swellings of long bones may be caused by infections, cysts or tumours. Spontaneous fractures may occasionally be the presenting symptom in the diagnosis of secondary carcinoma, multiple myeloma or generalized osteitis fibrosa cystica (hyperparathyroidism) or osteogenesis imperfecta.

Palpation

On palpation, bone tenderness occurs in local lesions when there is destruction, elevation or irritation of the periosteum, as in generalized osteitis fibrosa cystica, myelomatosis, bone infections, occasionally in carcinomatosis of bones and, rarely, in leukaemia. Injury is the most common cause.

Fractures

Fractures are common and may involve any bone. They are painful, distressing for the patient and expensive for the community (Box 15.9). Fractures in healthy bones commonly involve the long bones and are usually caused by trauma. Fractures of the wrists, hips and vertebrae are more frequently complications of bone disease, such as osteoporosis. Multiple rib fractures, caused by falls, may be found in heavy alcohol users but may only be seen as healed lesions on chest X-ray. Fractures occur without apparent trauma when a bone is weakened by disease, especially with metastatic malignant deposits in bone (pathological fractures). Traumatic fractures invariably present with local pain, swelling and loss of function, but pathological fractures may be relatively silent. The history will reveal the circumstances of the trauma, whether accidental or owing to physical abuse, and should be carefully documented, if necessary, with diagrams or digital photographs of the clinical findings. If clinical photographs are taken, prior written consent is needed.

Box 15.9 Fractures: clinical features

Type
- Closed
- Compound (open)

Complications
- Accompanying soft-tissue injury (indirect)

Features
- Haemorrhage
- Deformity
- Pain
- Crepitus
- Restricted movement

Cause
- Traumatic
- Spontaneous (osteoporosis or metabolic)
- Pathological

Examination of suspected fracture

Fractures may be open (compound) or closed. In closed fractures, the surrounding soft tissues are intact. In open fractures, the bone communicates with the surface of the skin, either because the primary injury has broken the overlying skin or because deformation at the fracture site has caused the bone ends to penetrate the skin. A major risk of infection exists when the fracture site communicates with the open air.

Deformity is an obvious feature in the majority of fractures, although in some patients this may be difficult to appreciate owing to a large body habitus. Comparison with the contralateral limb is often helpful. Deformity may be clinically characteristic, as in Colles' fracture, in which there is a fracture of the distal end of the radius characterized by dorsal displacement and angulation, shortening of the wrist and rotation of the fragment, well summarized in the description 'dinner-fork deformity'. Certain fractures may show little deformity; for example, a fracture of the femoral shaft may be accompanied by only slight deformity, because there is often little separation of the bones at the fracture site and other features are disguised by the thick overlying muscle. A fractured neck of the femur causes deformity through external rotation of the foot and shortening of the leg.

Most fractures are characterized by local tenderness and swelling, unless the overlying muscle mass is large, such as at the hip. Bony crepitus, owing to abnormal motion at the fracture site, is a feature of fractures, but this should not be elicited unless absolutely necessary for diagnosis because it is very painful. However, if it is perceived or has been recognized by the patient, it is diagnostic.

A fracture of the bone may damage the neighbouring soft tissues directly or, alternatively, a



Figure 15.21 Symmetrical joint involvement caused by rheumatoid arthritis.



Figure 15.22 Gouty tophus of the index finger.



Figure 15.23 Acute synovitis of the interphalangeal joint in reactive arthritis of the left hallux. Differential diagnosis includes gout.

fracture may be a marker of severe injury in which direct damage to the soft tissues, such as the nerves and vessels, may have taken place. A neuromuscular assessment distal to the fracture site is imperative. The presence or absence of pulses and cutaneous sensation and the colour and perfusion of the limb must always be recorded and any changes over time reported. Limb ischaemia distal to a fracture is a surgical emergency. Voluntary movement at joints distal to a fractured long bone, such as ankle movement in a fractured femur, should be noted. If this is absent, nerve injury must be suspected.



Figure 15.24 Nodal osteoarthritis (Heberden's nodes).

The joints

Examination of the joints can be summarized simply as 'look, feel and move' (i.e. inspection, palpation and range of movement). With practice, the clinician can develop a systematic review of the joints; for example, the jaw, cervical spine, shoulder girdle and upper limb, thoracic and lumbar spine, pelvis and lower limb, so that inconspicuous but important joints, such as the temporomandibular, sternoclavicular and sacroiliac joints, will not be overlooked. Compare the corresponding joints on the two sides of the body and always take care to avoid causing undue discomfort.

Inspection

The detection of joint inflammation is a crucial clinical skill. Inflammation is often associated with redness of the joint and with tenderness and warmth. Look also for swelling or deformity of the joint. The overall pattern of joint involvement should be recorded. Examine the contralateral joint where possible and note whether the distribution is symmetrical, as is usual in rheumatoid arthritis (Fig. 15.21), or asymmetrical, as in psoriatic arthropathy or gout (Fig. 15.22). The seronegative spondyloarthropathies (Fig. 15.23) tend to involve predominantly the joints of the lower limb.

Palpation

On palpation of a joint, check first for tenderness. Then determine whether the swelling is owing to bony enlargement or osteophytes (e.g. Heberden's nodes, Fig. 15.24), thickening of synovial tissues, such as occurs in inflammatory arthritis, or effusion into the joint space. Joint effusions usually have a characteristically smooth outline and fluctuation is usually easily demonstrable. Palpation of the whole joint is important because this may reveal localized enthesitis (e.g. lateral epicondylitis at the elbow) or reveal localized effusions (e.g. Baker's cyst in the posterior aspect of the knee joint). Tenderness and

Box 15.10 Assessment of joint tenderness

- Grade 1: The patient says the joint is tender.
- Grade 2: The patient winces.
- Grade 3: The patient winces and withdraws the affected part.
- Grade 4: The patient will not allow the joint to be touched.

Box 15.11 Extra-articular features of joint disease

- Cutaneous nodules
- Cutaneous vasculitic lesions
- Lymphadenopathy
- Oedema
- Tendon sheath effusions
- Enlarged bursae
- Ocular inflammation
- Diarrhoea
- Urethritis
- Orogenital ulceration
- Alopecia

enlargement of the ends of long bones, particularly the radius, ulna and tibia, can occur in hypertrophic pulmonary osteoarthropathy; a chest X-ray is essential. Gross disorganization of a joint—nearly always the foot and ankle joints—associated with an absence of deep pain and position sense occurs in neuropathic (Charcot) joints. Charcot joints probably arise from recurrent painless injury and overstretching and are a feature of severe chronic sensorimotor neuropathy.

Joint tenderness may be graded depending on the patient's reaction to firm pressure of the joint between finger and thumb (Box 15.10). Grade 4 tenderness occurs only in septic arthritis, crystal arthritis and rheumatic fever.

If tenderness is present, localize it as accurately as possible and determine whether it arises in the joint or in neighbouring structures, for example in the supraspinatus or bicipital tendon rather than the shoulder joint. Other rheumatic conditions produce different types of pain. For example, in conditions such as complex regional pain syndrome (CRPS) type I (previously known as reflex sympathetic dystrophy), hyperalgesia (abnormal and excessive pain to mildly painful stimuli) and allodynia (a sensation of pain to stimuli not normally painful, e.g. light touch) may often be seen, together with altered sweating and discolouration. In fibromyalgia, there is widespread diffuse muscle and joint tenderness but no inflammation. A number of characteristic trigger points may be tender around the neck, trunk and upper and lower limbs.

Tendon sheath crepitus

Tendon sheath crepitus is a grating or creaking sensation defined by palpating the tendon while the patient is asked to contract the muscle tendon complex involved. It is particularly common in the hand and is seen in rheumatoid arthritis and systemic sclerosis. In tenosynovitis of the long flexor tendons in the palm, tendon sheath crepitus may be associated with the trigger phenomenon, in which the finger becomes caught in flexion and has to be pulled back into extension. Tendon sheath effusions can be distinguished from joint swelling by their anatomical location in association with tendons; where there is doubt, an ultrasonographic assessment will provide clarity.

Box 15.12 Types of subcutaneous nodules

- Gouty tophi caused by urate deposition
- Rheumatoid nodules
- Vasculitic nodules in systemic lupus erythematosus (SLE) and systemic vasculitis
- Xanthomatous deposits (hypercholesterolaemia)



Figure 15.25 Rheumatoid nodule overlying the olecranon of the right arm.

Joint crepitus

Joint crepitus can be detected by feeling the joint with one hand while moving it passively with the other. This may indicate osteoarthritis or loose bodies (cartilaginous fragments) in the joint space, but should be differentiated from non-specific clicking of joints.

Range of movement

When examining joints for range of movement, it is usually sufficient to estimate the degree of limitation based on comparison with the normal side or on the examiner's previous experience. For an accurate

description, the actual range of movement should be measured with a protractor (goniometer). Both active and passive movement should be assessed. Active movement, however, may give a poor estimation of the true range of movement because of muscle spasm owing to pain. If pain is very severe on attempting active movement and other findings suggest a fracture, take an X-ray before attempting any further examination. In testing the range of passive movement, always be gentle, particularly when the joints are painful. The direction of joint movement, particularly in ball and socket joints, may provide additional information as to the underlying lesion.

Limitation of movement in a joint may be owing to pain, muscle spasm, contracture, inflammation, increased thickness of the capsular or periarticular structures, effusion into the joint space, bony overgrowths, bony ankylosis, mechanical factors such as a torn meniscus or to painful conditions quite unconnected with the joint.



Figure 15.26 Nail-fold vasculitis (**A**) A patient with dermatomyositis demonstrating nailfold infarcts at the cuticle and evidence of Gottrons papules on the dorsum of the MCP joints. (**B**) Nailfold infarcts seen in a patient with SLE.

Extra-articular features of joint disease

Some of the extra-articular features of joint disease are listed in Box 15.11.

Subcutaneous nodules

Subcutaneous nodules are associated with several conditions (Box 15.12). If gout is suspected, inspect the helix of the ear for tophi caused by the subcutaneous deposition of urate, which may also be found overlying joints or in the finger pads (see Figs 15.15 and 15.22). Subcutaneous nodules in rheumatoid arthritis are firm and non-tender; they may be detected by running the examining thumb from the point of the elbow down the proximal portion of the ulna (Fig. 15.25). They can also be found at other pressure and frictional sites, such as bony prominences, including the sacrum and in the hands. If an olecranon bursa swelling is found, feel also in its wall, as rheumatoid nodules, tophi or occasionally xanthomata may be found within the swelling. Subcutaneous nodules are not specific to rheumatoid arthritis and may occur in patients with SLE, systemic sclerosis and rheumatic fever. Ultrasonographic appearance may be helpful in distinguishing subcutaneous nodules and may facilitate a guide aspiration if necessary.

Cutaneous vasculitic lesions

Cutaneous vasculitic lesions may be seen in rheumatoid arthritis, SLE, Sjögren's syndrome and the systemic vasculitides. They may be small, punched-out necrotic lesions, palpable purpura or vasculitic ulcers, especially on the lower limbs. Small vessel involvement is typically seen at the nail fold (cutaneous infarct; Fig. 15.26), but it also occurs at pressure sites. Splinter haemorrhages, the classic feature of bacterial endocarditis, may also be a feature of vasculitis and the antiphospholipid syndrome.

Skin

Several rheumatological conditions occur with skin manifestations. Vasculitic skin lesions have already been mentioned earlier, but there may be other clues to possible underlying rheumatic conditions which should be actively sought during examination.

Skin psoriasis is a useful indication of possible psoriatic arthritis, particularly if there is associated nail pitting (Fig.15.27). Psoriasis is often noted on



Figure 15.27 Psoriartic arthritis. The patient has swelling of the distal interphalangeal joint (DIP) on the overlying right hand and third DIP joint of the left hand. Nail dystrophy with pitting is seen.

the extensor parts of the elbows and knees, but may be seen in the hairline, at the umbilicus, behind the ears as well as in the natal cleft.

Conditions, such as dermatomyositis, may present with the predominant dermatological manifestations such as peri-orbital oedema and purplish discolouration (Fig. 15.28). Gottrons papules are often noted on extensor surfaces at the level of the metacarpophalangeal joints (Fig.15.26).

Skin ulceration can also be seen in the context of vasculitis and typically are well-defined punched lesions, e.g. Felty's syndrome.

Lymphadenopathy

Lymphadenopathy may be found proximal to an inflamed joint, not only in septic arthritis but also in rheumatoid arthritis. Generalized lymphadenopathy, sometimes with splenomegaly, is common in active SLE.

Local oedema

Local oedema is sometimes seen over inflamed joints (Fig. 15.29), but other causes of oedema must be excluded. Pitting leg oedema may indicate cardiac failure, pericardial effusion or nephrotic syndrome, which can complicate rheumatoid arthritis and SLE.

Other soft-tissue swellings

Tendon sheath effusions are distinguished from joint swellings by their location in association with tendons. Enlarged subcutaneous bursae may be found over pressure areas, particularly at the olecranon surface of the elbow, owing to inflammatory joint disease or secondary to friction. Deeper bursae may be defined only by finding local tenderness or by stressing adjacent tissues (e.g. greater trochanter bursitis).



Figure 15.28 Patient with dermatomyositis presenting with a typical heliotrope rash around the eyes.



Figure 15.29 Pitting oedema, right hand.

Examination of individual joints

The range of movement of joints is described in the following pages. All motion should be measured in degrees from a neutral or zero position, which must be defined whenever possible and compared with the opposite side. Some special features seen at individual joints are set out in each section.

The spine

General examination of the vertebral column

Inspection

Examine the patient both in standing and in sitting in the erect posture. The normal thoracolumbar spine has an S-shaped curve. If there is an abnormality, note which vertebrae are involved and at what level any vertebral projection is most prominent. Note the presence of any local projections or angular deformity of the spine. Torticollis, if present, is usually obvious, resulting from muscle spasm of



Figure 15.30 Scoliosis of the lumbar spine owing to a prolapsed intervertebral disc.

the sternocleidomastoid, trapezius and other neck muscles. This leads to the neck being in a tilted, rotated, partially flexed position.

Palpation

The major landmarks are the spinous processes of C7 (the vertebra prominens) and the last rib, which articulates with the 12th thoracic vertebra. In many patients, however, the last rib cannot be felt distinctly, which therefore makes this rather untrustworthy as a guide to this level.

The neutral position of the spine is an upright stance with the head erect and the chin drawn in. Note any curvature of the spinal column, whether as a whole or a part of it. Abnormal curvature may be in an anterior, posterior or lateral direction (Fig. 15.30). Anterior curvature (convex forward) is termed lordosis. There are natural lordotic curves in the cervical and lumbar regions. Posterior curvature (convex backward) is termed kyphosis. The thoracic spine usually exhibits a slight smooth kyphosis, which increases in the elderly and especially in osteoporosis. It must be distinguished from a localized angular deformity (gibbus, Fig. 15.31) caused by a fracture, by Pott's disease (spinal tuberculosis, Fig. 15.32) or by a metastatic malignant deposit.



Figure 15.31 Gibbus of the lumbar spine caused by tuberculosis.



Figure 15.32 X-ray of tuberculous discitis. This shows the underlying deformity shown in Figure 15.31. There is tuberculous infection of the intervertebral disc, causing the spinal deformity.

Lateral curvature is termed scoliosis (see Fig. 15.30) and may be towards either side. It is always accompanied by rotation of the bodies of the vertebrae in such a way that the posterior spinous processes come to point towards the concavity of the curve. The curvature is always greater than appears from inspection of the posterior spinous processes. In scoliosis owing to muscle spasm (e.g. with lumbosacral disc protrusion syndromes), the spinal curvature and rotational deformity decrease in flexion. When scoliosis is caused by inequality

of leg length, it disappears on sitting, because the buttocks then become level. Scoliosis secondary to skeletal anomalies shows in spinal flexion as a 'rib hump' owing to the rotation. Kyphosis and scoliosis are often combined, particularly when the cause is an idiopathic spinal curvature, beginning in adolescence.

The cervical spine

The following movements should be tested (Fig. 15.33):

- Rotation: ask the patient to look over one then the other shoulder.
- Flexion: ask the patient to touch chin to chest.
- Extension: ask the patient to look up to the ceiling.
- Lateral bending: ask the patient to bend the neck sideways and to try to touch the shoulder with the ear without raising the shoulder.

Note any pain or paraesthesiae in the arm reproduced by neck movement, especially on gentle sustained extension or lateral flexion, suggesting nerve-root involvement. If indicated, check for any associated neurological deficit, particularly of radicular or spinal cord type.

In rheumatoid arthritis, particular care is necessary when examining the neck, as atlantoaxial instability may lead to damage to the spinal cord when the neck is flexed. If there is any doubt about neck stability in a patient with rheumatoid arthritis, arrange for lateral X-rays of the cervical spine in flexion and extension, together with a view of the odontoid peg through the mouth, and defer clinical examination.

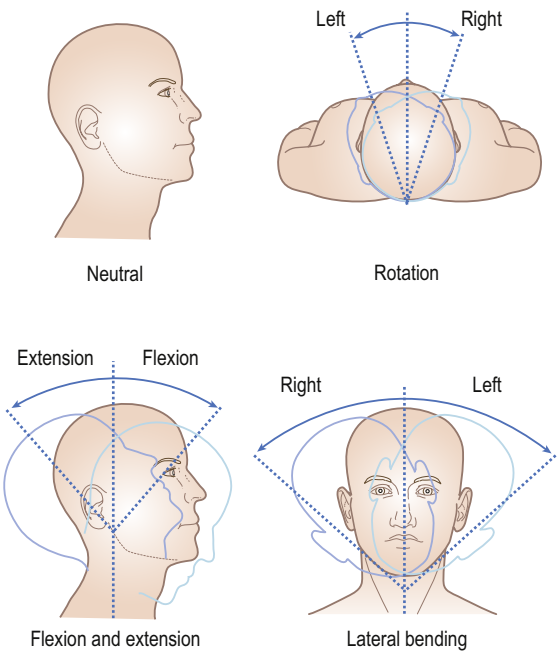


Figure 15.33 Movements of the neck.

In patients with cervical injury, never try to elicit range of motion of the neck. Instead, splint the neck, take a history, look for abnormality of posture (usually in rotation) and check neurological function in the limbs, including both arms and both legs. Imaging the neck in the lateral (Fig. 15.34) and anteroposterior planes, *without* moving the neck, is essential and may be done with plain radiography or, in trauma patients, with computed tomography or magnetic resonance imaging (MRI) to assess injury to the cord. Only if imaging is normal should neck movements be examined.

The thoracic and lumbar spine

The main movement at the thoracic spine is rotation, whereas the lumbar spine can flex, extend and bend laterally. The following movements should be tested (Fig. 15.35):

- Flexion: ask the patient to try to touch his toes, without bending at the knees.
- Extension: ask the patient to bend backwards.
- Lateral bending: ask the patient to run the hand down the side of the thigh as far as possible.
- Thoracic rotation: ask the seated patient, with arms crossed, to twist round to the left and right as far as possible.

In flexion, the normal lumbar lordosis should be abolished. It is important to distinguish between flexion at the lumbar spine and compensated flexion



Figure 15.34 Lateral X-ray of cervical spine showing degenerative spondylosis with narrowing of the disc spaces and reversed cervical lordosis between C4 and C6.

at the hips, which may deceive the examiner if only a cursory examination is carried out. A more objective assessment of lumbar spine flexion can be performed by marking a vertical 10-cm line on the skin overlying the lumbar spinous processes and the sacral dimples and measuring the increase in the line length on flexion (modified Schober's test); this should normally be 5 cm or more. Painful restriction of spinal movement is an important sign of cervical and lumbar spondylosis, but it may also be found in vertebral disc disease or other mechanical disorders of the back or neck in association with muscle spasm. A useful clinical aphorism is that a rigid lumbar spine should always be investigated for serious pathology, such as infection (e.g. staphylococcal or tuberculous discitis), malignancy or inflammation (e.g. ankylosing spondylitis). Spinal movements may be virtually absent in ankylosing spondylitis (Fig. 15.36), but in the early stages of this condition lateral flexion of the lumbar spine is typically affected first. In mechanical or osteoarthritic back problems, flexion and extension are reduced more than lateral movements. In prolapsed intervertebral disc lesions, sustained gentle lumbar extension may reproduce the low back pain and sciatic radiation.

Chest expansion is a measure of costovertebral movement and should be recorded using a tape measure with the patient's hands behind his head to reduce the possibility of muscular action in the shoulder girdle giving a false reading. The examiner stands behind the patient and the tape measure is placed around the chest at the level of the xiphisternal joint. The patient is asked to exhale and the tape is tightened before a maximum inhalation. An increase of 5 cm is expected in adults. Reduced chest expansion is a characteristic early feature in ankylosing spondylitis. Obviously, this may also be a feature of primary pulmonary disease, such as emphysema.

Examination of the back is completed by assessing straight leg raising (SLR) and strength, sensation and reflex activity in the legs. Pain and limitation on SLR are features of a prolapsed intervertebral disc when there is irritation or compression of one of the roots of the sciatic nerve. Tight hamstring muscles may cause a similar picture, but if there is severe pain, it is more considerate to lower the leg to just below the limit of SLR and then to see whether gentle passive dorsiflexion of the foot brings back the same pain. If in doubt, dorsiflex the foot once the limit of SLR has been reached. This further stretches the sciatic nerve (the pain increases) but does not affect the hamstrings (Lasègue's sign). The femoral stretch test is a useful confirmatory test. It is performed with the patient lying prone: if there is a prolapsed disc at that level, flexion at the knee will produce

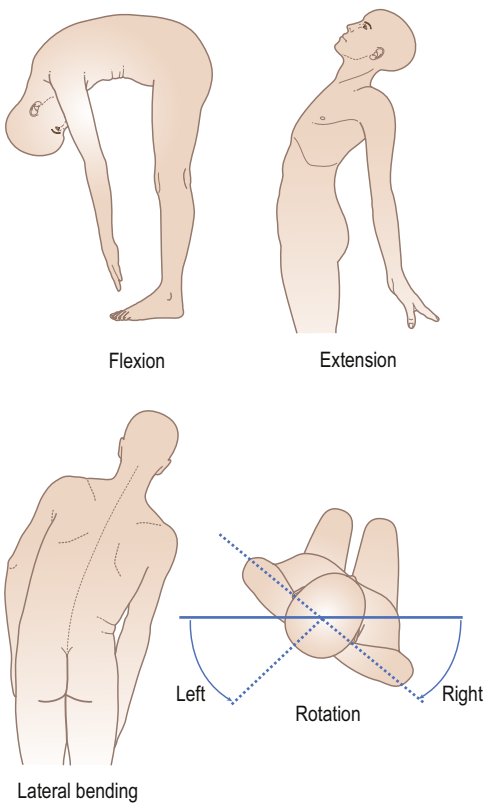


Figure 15.35 Movements of the lumbar and dorsal spine.



Figure 15.36 Ankylosing spondylitis. Note dorsal kyphosis and protuberant abdomen owing to poor chest expansion with abdominal breathing.

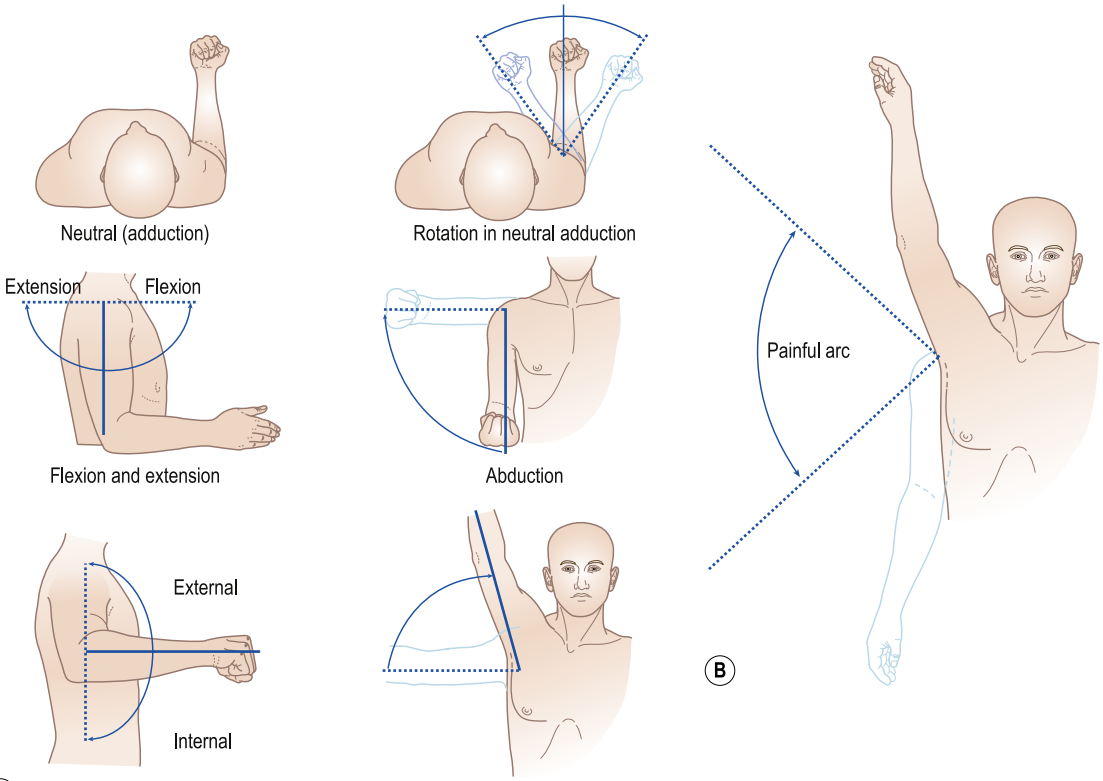


Figure 15.37 **(A)** Movements of the shoulder. **(B)** Painful arc of supraspinatus tendinitis.

pain in the lower lumbar spine. Sacral sensory loss must always be carefully assessed because, if there is a large central lumbosacral disc protrusion, bilateral limitation of SLR may be associated with bladder or bowel dysfunction and sacral anaesthesia. This combination is an emergency and requires immediate investigation and treatment.

The sacroiliac joints

The surface markings of these joints are two dimples low in the lumbar region. Test for irritability in the following ways:

- Direct pressure over each sacroiliac joint
- Firm pressure with the side of the hand over the sacrum
- Inward pressure over both iliac bones with the patient lying on one side, in an attempt to distort the pelvis
- Flex the hip to 90° and exert firm pressure at the knee through the femoral shaft (this should only be done if the hip and knee are not painful).

In the last three, a positive test is only indicated by the patient localizing discomfort to the sacroiliac joint.

The shoulder

Shoulder examination involves the assessment of a number of other joints in addition to the glenohumeral articulation. Abduction of the shoulder particularly involves movement at the sternoclavicular, acromioclavicular, glenohumeral and scapulothoracic joints. Pain is also referred and the shoulder examination should be accompanied by an examination of the cervical spine, because radicular pain may be localized by the patient at the shoulder and vice versa.

Inspection of the shoulder should be from the posterior, lateral and anterior positions. Where possible, the contralateral shoulder should be exposed and comparisons made in muscle bulk.

Palpation of the shoulder should include the long head of the biceps tendon in the bicipital groove, acromioclavicular joint, clavicle, sternoclavicular joint and the acromion.

The neutral position is with the arm to the side, elbow flexed to 90° and forearm pointing forwards. Because the scapula is mobile, true shoulder (glenohumeral) movement can be assessed only when the examiner anchors the inferior angle of the scapula between finger and thumb on the posterior chest wall. The following movements should be tested (Fig. 15.37):

- Flexion
- Extension
- Abduction
- Rotation in abduction
- Rotation in neutral position
- Elevation (also involving scapular movement)

In practice, internal rotation can be compared best by recording the height reached by each thumb up the back, representing combined glenohumeral and scapular movement. Similarly, external rotation can be assessed by the ability to get the hand to the back of the neck. Limitation of external rotation is a good sign of true glenohumeral disease, which may occur in adhesive capsulitis (frozen shoulder) or erosive damage from inflammatory arthritis.

Note any pain during the range of movement. In supraspinatus tendinitis, a full passive range of movement is found, but there is a painful arc on abduction, with pain exacerbated on resisted abduction (see Fig. 15.37). Other tendon involvement should also be defined by pain on resisted action.

Subacromial impingement owing to a bursitis or rotator cuff abnormality may produce severe pain at the end of abduction, blocking full elevation. Acute bursitis, however, may be so painful that no abduction is allowed (grade 4 discomfort). Acromioclavicular joint pain is always very localized and is typically felt in the last 10° of elevation (170° to 180° arc).

Special tests, as follows, are used in shoulder examination.

Supraspinatus

The supraspinatus muscle can be tested using the 'empty can' test, which involves forward flexion of the arm to 90° and abduction to 30° with the thumb pointing upwards. The shoulder is then internally rotated so that the thumb is now facing downwards. The examiner places his hand on the arm and exerts downward pressure while the patient resists. Weakness or significant pain may indicate a muscle tear and further imaging is recommended (ultrasound (US) or MRI). It is recommended that both arms be used in this test for comparison purposes.

Subscapularis

The patient is asked to place his hand behind the back with the dorsum of the hand resting in the region of the mid-lumbar spine (mainly internal rotation at the glenohumeral joint). The ability actively to lift the dorsum of the hand off the back constitutes a normal test with no dysfunction or rupture of subscapularis.

Infraspinatus and teres minor

Hornblower's sign is an inability to rotate the elevated arm externally. This movement indicates severe infraspinatus and teres minor weakness.

Long head of biceps

Hueter sign may indicate a ruptured tendon. The patient is seated with the elbow extended and forearm supinated. The elbow is then flexed by the patient against resistance. If the tendon is ruptured, a biceps 'ball' develops.

Yergason's test indicates if there is bicipital tendinitis. The patient's elbow is flexed and the forearm pronated. The examiner holds the arm at the wrist while the patient actively supinates against resistance. If this resisted movement produces pain located to the bicipital groove area, then pathology within the long head of the biceps tendon sheath is likely.

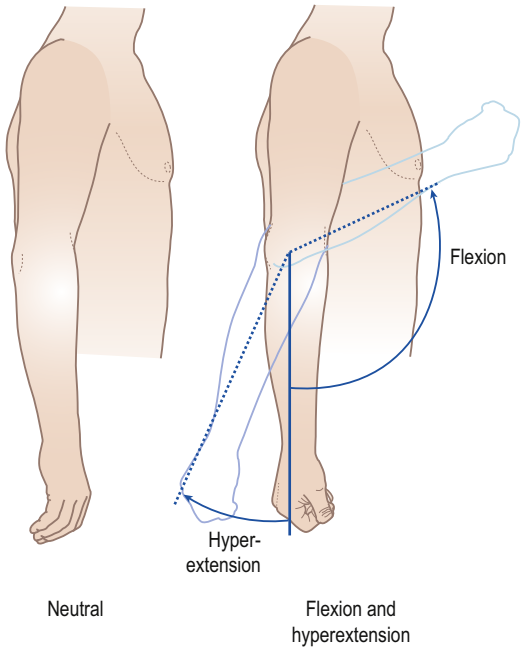


Figure 15.38 Movements of the elbow.

The elbow

The neutral position is with the forearm in extension. The following movements should be tested (Fig. 15.38) as follows:

- Flexion
- Hyperextension

Medial (golfer's elbow) and lateral (tennis elbow) epicondylitis are the most common causes of elbow pain. They are characterized by pain on active use but, if severe, may be associated with night pain.

Examination must define localized epicondylar tenderness with pain on resisted movement. Wrist extension exacerbates lateral epicondylar tenderness and wrist flexion exacerbates medial epicondylar tenderness. An elbow effusion may be palpated in the posterior triangle formed by the epicondyles and the olecranon.

Placing the thumb and second and third fingers on the lateral, medial epicondyles and olecranon, respectively, with the joint in an extended position produces a straight alignment. In a flexed position

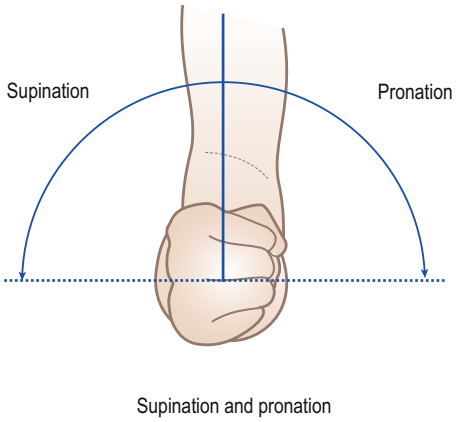


Figure 15.39 Movements of the forearm.

Figure 15.40 Movements of the wrist.

these three points form a triangular shape. Disturbance of this geometric shift may indicate a supracondylar fracture.

The forearm

The neutral position is with the arm by the side, elbow flexed to 90° and thumb uppermost. The following movements should be tested (Fig. 15.39):

- Supination
- Pronation

The wrist

Inspection of the wrist may reveal clear evidence of joint swelling or localized tenosynovitis. The distal ulna is usually visible as a smooth, rounded protrusion. Loss of the normal anatomical landmarks may suggest joint or tendon inflammation.

Palpation of the wrist for warmth and swelling should be performed as well as specific palpation for tendon and bone discomfort. Palpation over the anatomical snuff box resulting in pain may indicate a scaphoid fracture if there is a history of trauma. Tenderness along the ulnar aspect of the wrist may suggest inflammation of the extensor carpi ulnas tendon, commonly involved in early rheumatoid arthritis. At the distal end of the anatomical snuff box the base of the thumb (first carpal-metacarpal joint) can be palpated, which is a commonly affected joint in generalized osteoarthritis in middle-aged and older patients.

The neutral position is with the hand in line with the forearm, and palm down. The following movements should be tested (Fig. 15.40):

- Dorsiflexion (extension)
- Palmar flexion
- Ulnar deviation
- Radial deviation

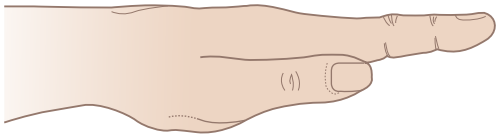
Even minor limitation of wrist flexion or extension can be detected by comparing movement in both wrists (Fig. 15.41). Limitation of the wrist joints is usually caused by inflammatory arthritis. Primary osteoarthritis of the wrist is rare, but secondary degenerative change is common.

The fingers

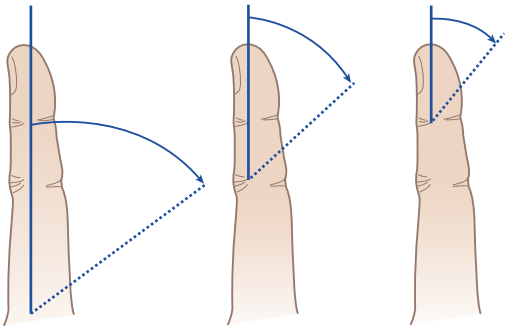
When identifying fingers, use the names thumb, index, middle, ring and little. Numbering tends to lead to confusion. The neutral position is with the fingers in extension. Test flexion at the metacarpophalangeal



Figure 15.41 Minor limitation of left wrist extension compared with the right. Note the slightly different angulation of the left forearm.



Neutral



Flexion of MCP | Proximal IP | Distal IP joints

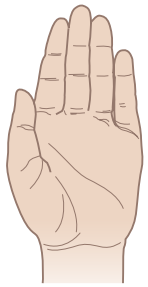
Figure 15.42 Movements of the fingers.

(MCP), proximal interphalangeal (PIP) and distal interphalangeal (DIP) joints (Fig. 15.42).

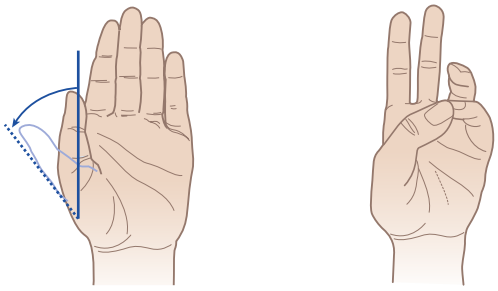
In fractures of the fingers, the most common deformity is rotational. If the finger will flex, make sure it points to the scaphoid tubercle (all the fingers will point individually in this direction). If it will not flex, look end-on at the nail and make sure it is parallel with its fellows.

The thumb (carpometacarpal joint)

The neutral position is with the thumb alongside the forefinger, and extended. The following movements should be tested (Fig. 15.43):



Neutral



Extension | Opposition

Figure 15.43 Movements of the thumb.

- Extension
- Flexion (measured as for the fingers)
- Opposition
- Abduction (not illustrated; movement at right angles to plane of palm)

The hand

Deformities in joint disease

Examination of the individual joints of the hand may be less informative than inspection of the hand as a whole (Fig. 15.44). The combination of Heberden's nodes and thumb carpometacarpal arthritis occurs in osteoarthritis (see Fig. 15.24). Pain and subluxation of the carpometacarpal joint is a typical feature of primary nodal osteoarthritis, leading to a 'square hand' appearance on making a fist.

A variety of patterns of deformity are characteristic of long-standing rheumatoid arthritis. For example, metacarpophalangeal joint subluxation, ulnar deviation of the fingers at the metacarpophalangeal joints, 'swan neck' (Fig. 15.45) and 'boutonnière' deformities (flexed proximal and hyperextended distal interphalangeal joints) of the fingers are typical in advanced disease. This is owing to the head of the phalanx sliding dorsally between the lateral slips of the extensor tendon, the middle slip having been damaged. In psoriatic arthritis, terminal interphalangeal joint swelling may occur, with psoriatic pitting and ridging of the nail (onychopathy) on that digit. Jaccoud's arthropathy



Figure 15.44 Functional ability. Severe joint deformity owing to psoriatic arthropathy, but retention of function and artistic ability.



Figure 15.45 Swan-neck deformity of the right hand. Note also wasting of the small muscles of the hand owing to disuse in this patient with rheumatoid arthritis.

produces a similar appearance to swan neck and boutonnière deformity alongside thumb subluxation and ulnar deviation. However, these deformities are reducible when the patient is asked to make a fist as a result of no underlying structural joint damage. This arthropathy may be seen in SLE and Sjögren's syndrome.

Deformities owing to neuropathy

The hand may adopt a posture typical of a nerve lesion (see Chapter 16). Slight hyperextension of the medial metacarpophalangeal joints with slight flexion of the interphalangeal joints is the 'ulnar claw hand' of an ulnar nerve lesion. There is wasting of the small muscles of the hypothenar eminence, with loss of sensation of the palmar and dorsal aspects of the little finger and of the ulnar half of the ring finger. In a median nerve lesion, the thenar eminence (abductor pollicis brevis) will be flattened (see Fig. 15.18) and sensory impairment will be found on the palmar surfaces of the thumb, index, middle and radial half of the ring fingers. Remember that carpal tunnel syndrome may be a presenting feature of wrist inflammation.

Box 15.13 Hip movements to be tested

- Flexion: measured with knee bent. Opposite thigh must remain in neutral position. Flex the knee as the hip flexes.
- Abduction: measured from a line that forms an angle of 90° with a line joining the anterior superior iliac spines.
- Adduction (measured in the same manner).
- Rotation in flexion.
- Rotation in extension.
- Extension: attempt to extend the hip with the patient lying in the lateral or prone position.

Assessment of hand function

Assessment of hand function (see Fig. 15.44) should include testing hand grip and pinch grip (between index and thumb). The latter may be decreased in lesions in the line of action of the thumb metacarpal, particularly scaphoid fractures.

The hip

The neutral position is with the hip in extension and the patella pointing forwards. Ensure the pelvis does not tilt by placing one hand over it while examining the hip with the other hand. Look for scars and wasting of the gluteal and thigh muscles. The hip joint is too deeply placed to be accessible to palpation. Hip movements to be tested are listed in Box 15.13 (Fig. 15.46).

Additional examination of the hip joint

- Test for flexion deformity. With one hand flat between the lumbar spine and the couch, flex the normal hip fully to the point of abolishing the lumbar lordosis. The spine will come down onto the hand, pressing it onto the couch. If there is a flexion deformity on the opposite side, the leg on that side will move into a flexed position (Thomas' test).
- Trendelenburg test. Observe the patient from behind and ask him to stand on one leg. In health, the pelvis tilts upwards on the side with the leg raised. When the weight-bearing hip is abnormal, owing to pain or subluxation, the pelvis sags downwards owing to weakness of the hip abductors on the affected side.
- Measurement of 'true' and 'apparent' shortening. The length of the legs is measured from the anterior superior iliac spine to the medial malleolus on the same side. Any difference is termed 'true' shortening and may result from disease of either the hip joint or the neck of the femur on the shorter side. 'Apparent' shortening is caused by tilting of the pelvis and can be measured by comparing the lengths of the two legs, measured from the umbilicus, provided there is no true shortening

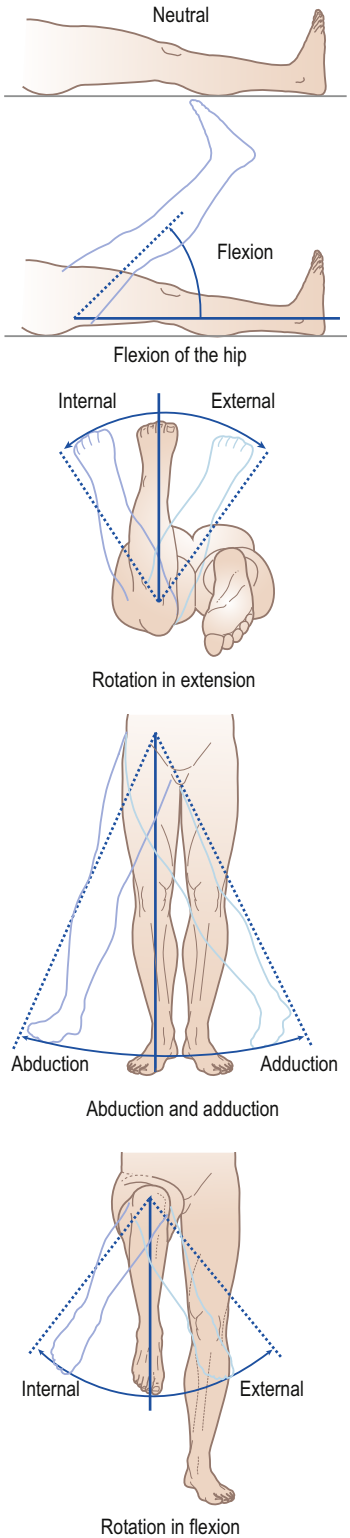


Figure 15.46 Movements of the hip.

of one leg. Apparent shortening is usually caused by an abduction deformity of the hip. Femoral or tibial shortening may be demonstrated with the patient lying on a couch and looking across both knees held equally flexed.

The knee

Magnetic resonance imaging scans of the normal knee are illustrated in Figure 15.47. The neutral position is complete extension. Observe any valgus (lateral angulation of the tibia) or varus (medial angulation) deformity on the couch and on standing. Look for muscle wasting. The quadriceps, especially the medial part near the knee, wastes rapidly in knee joint disease. Swelling may be obvious, particularly if it distends the suprapatellar pouch. Check the apparent height of the patella and watch to see if it deviates to one side in flexion or extension of the knee. Feel for tenderness at the joint margins, not forgetting the patellofemoral joint. Palpate the ligaments, remembering that the medial collateral ligament is attached 8 cm below the joint line. Measure the girth of the thigh muscles 10 cm above the upper pole of the patella.

Joint swelling

The presence of swelling in the knee joint may be confirmed by the patellar tap test or, for small effusions, by the bulge test, in which the medial parapatellar fossa is emptied by pressure of the flat of the hand sweeping proximally. The bulge is seen to refill as the suprapatellar area is emptied by pressure from the flat hand. Posterior knee joint (Baker's) cysts, particularly in rheumatoid arthritis, may be palpable in the popliteal fossa. They sometimes rupture, producing calf pain, and may then mimic a deep vein thrombosis. When intact, large posterior knee cysts can sometimes cause venous obstruction.

The movements of the knee are flexion and extension (Fig. 15.48). Loss of flexion can be documented by loss of the angle of flexion or loss of heel-to-buttock distance, either in the crouching position or on the couch. Loss of extension is detected by the inability to get the back of the knee onto the flat examining couch. Hyperextension must be sought by lifting the foot with the knee extended and comparing it with the normal side. Lack of full extension by comparison with the normal constitutes fixed flexion deformity. Loose bodies in the joint cause crepitus, interruption of movement (locking) and pain and effusion (Fig. 15.49).

Testing for stability

Cruciate ligaments

Anterior/posterior drawer test. The patient should be situated on the couch with the knee placed in 90° flexion. If there is a posterior sag on inspection, this may provide a false-positive anterior drawer sign. Place both hands around the upper part of the leg with both thumbs placed on the tibial tuberosity and fingers around the posterior aspect of the knee. With your forearm resting on the shin, firmly pull

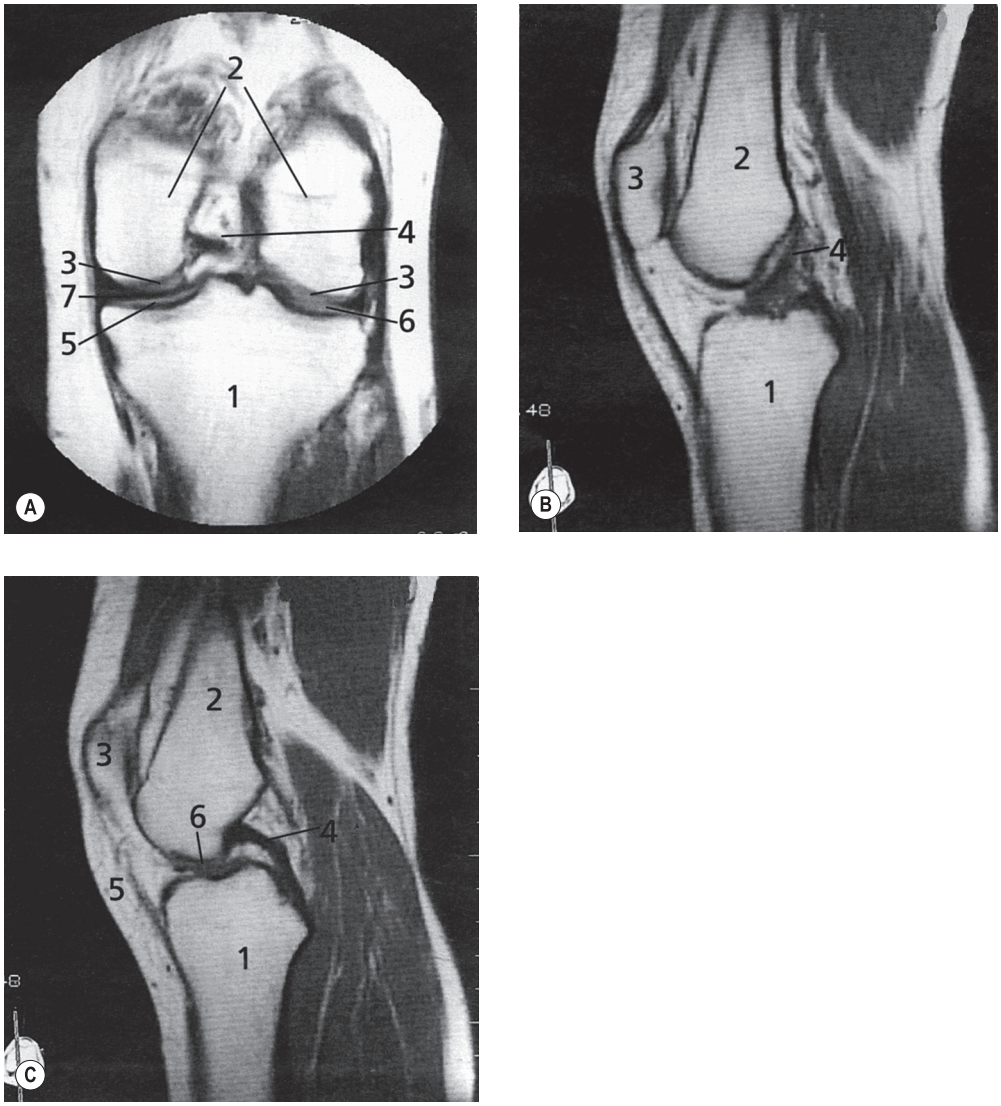


Figure 15.47 Magnetic resonance imaging (MRI) scans of the normal knee (MRI, T1-weighted). **(A)** Scan to show the medial (5) and lateral (6) menisci, origin of the anterior cruciate ligament (4) and the articular cartilages (3) and synovial fluid (7). Other structures shown are (1) tibia, (2) articular surfaces of femur. **(B)** The anterior cruciate ligament (4). Other structures shown are (1) tibia, (2) femur, (3) patella. **(C)** The posterior cruciate ligament (4). Other structures shown are (1) tibia, (2) femur, (3) patella, (5) patellar tendon, (6) joint space.

the tibia anterior. A significant forward movement of the tibia plateau suggests an anterior cruciate tear. Pushing backwards with posterior movement of the tibial plateau may suggest a posterior cruciate pathology.

Collateral ligaments

The collateral ligaments are best assessed with the knee in 30° of flexion. Place your right hand on the lower limb (medial mid-calf) and left hand on the upper lateral thigh. Gradually apply opposing force to detect excessive medial movement of the lower limb. This tests the integrity of the medial collateral ligament. Reverse hand positions (left hand on lateral mid-calf and right hand on medial lower thigh) and repeat the movement to test the lateral collateral ligament. Pain on testing indicates possible enthesitis or tear. Excessive movement can be seen in the context of joint hypermobility, which should be considered as part of the overall examination.

The ankle

The ankle is a hinge joint with movement only in the sagittal plane. The neutral position is with the outer border of the foot at an angle of 90° with the leg and midway between inversion and eversion. Observe the patient from behind in the standing position. With any long-standing ankle disorder, there will be a loss of calf muscle bulk.

Look at the position of the foot with the patient standing. The heel may tilt outwards (valgus deformity) in subtalar joint damage. Inward (varus

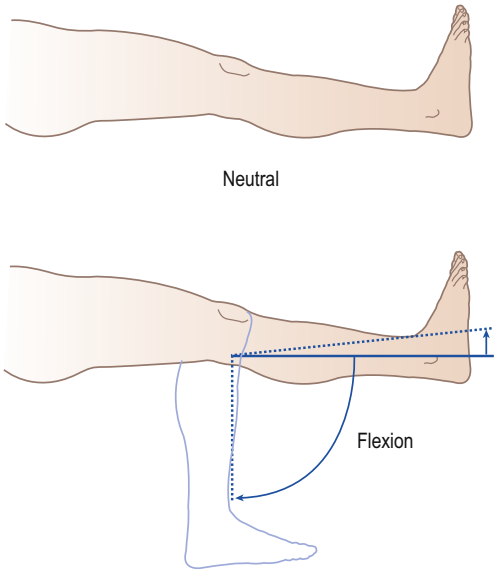


Figure 15.48 Movements of the knee.

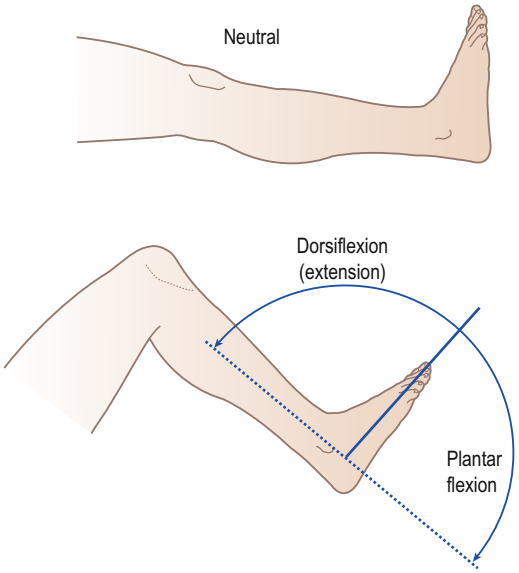


Figure 15.50 Movements of the ankle.

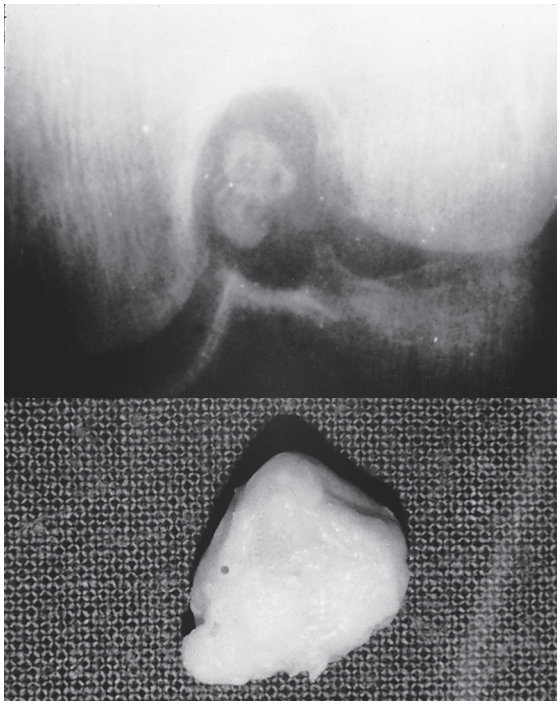


Figure 15.49 Loose body in tunnel view X-ray of knee, showing the loose body in the intercondylar space.



Figure 15.51 Daylight sign owing to metatarsophalangeal joint synovitis in rheumatoid arthritis.

deformity) is much less common and usually not so painful. Flattening of the longitudinal arch of the foot (pes planus) also produces valgus at the heel, but the foot curves laterally as well because the change is in the mid-tarsal joints in addition to the subtalar joint.

The following movements should be tested (Fig. 15.50):

- Dorsiflexion: test with the knee in flexion and extension to exclude tight calf muscles.
- Plantarflexion: place a finger on the head of the talus to be sure that it is moving. A hypermobile subtalar joint can mimic movement in an arthrodesed ankle.

The foot

Remember that complaints apparently relating to the foot may be features of systemic disease, such as gout, or of referred vertebral problems, such as a prolapsed intervertebral disc. Look for abnormalities of posture.

Callosities are areas of hard skin under points of abnormal pressure. The most common site is beneath the metatarsal heads because loss of the normal soft tissue pad allows abnormal loading. There may be abnormal spread of two adjacent toes (daylight sign: Fig. 15.51) on weight-bearing if there

is synovitis between the metatarsal heads. Check for lateral deviation of the big toe (hallux valgus), usually associated with abnormal swelling at its base (a bunion). There may be deformities affecting any or all toes, with abnormal curvature (claw toes), fixed flexion of the terminal joint (hammer toes) or overriding.

The foot consists of three regions: hindfoot (subtalar joint), mid-foot (tarsal joints) and forefoot (MTP, PIP, DIP joints). The following movements should be tested (Fig. 15.52):

- Subtalar inversion and eversion: cup the heel in the hands and move it in relation to the tibia without any up and down movement; this eliminates movement at the ankle or mid-tarsal joints.
- Mid-tarsal inversion/eversion and adduction/abduction: hold the os calcis in the neutral position in one hand and grasp and rotate the forefoot in the other.
- Metatarsophalangeal and interphalangeal flexion/extension.

Also look for tenderness or swelling at the Achilles tendon insertion on the back of the calcaneum and for plantar tenderness at the site of the plantar fascial insertion. Inflammation of these attachments (enthesopathy) is common in ankylosing spondylitis and other spondyloarthropathies.

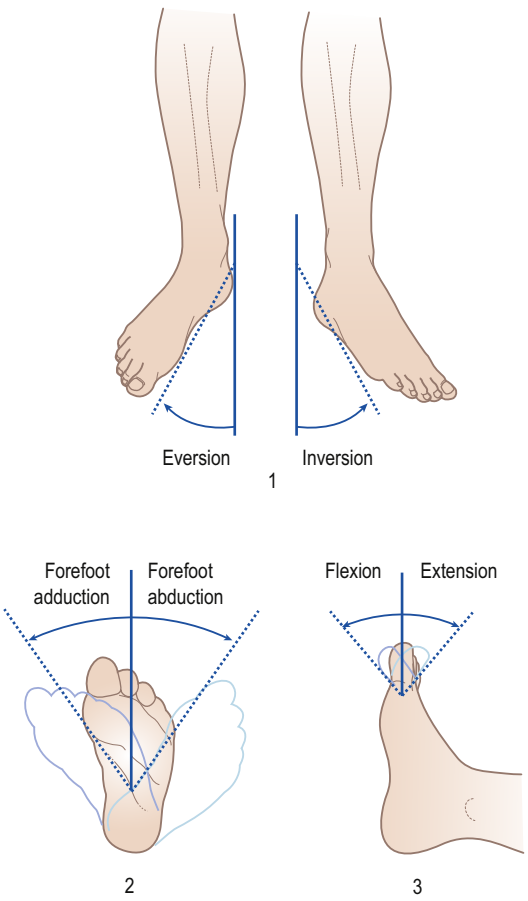


Figure 15.52 Movements of the foot.

Achilles tendon rupture

Thompson test: This test requires the patient to be in the prone position with the feet hanging off the table. The clinician then squeezes the patient's calf muscle, which should shorten the Achilles tendon causing the ankle to plantar flex. It is important to note that false-negative results may occur in older injuries, where organization of a hematoma can cause some reconstitution of the tendon or with an incomplete partial tear. Patients may still be able to walk and actively plantar flex the ankle despite significant Achilles tendon tear. Achilles rupture is most often seen in the context of sport or with a clear history of trauma but may also occur in patients with spondyloarthropathies. When an index of suspicion of an Achilles tear or rupture arises, imaging with ultrasound or MRI is recommended.

The gait

It is best to study gait with the patient's legs and feet fully exposed and *without* socks, shoes or slippers. Ask the patient to walk away from you, to turn around at a given point and then to walk towards you.

Abnormalities of gait usually are caused either by joint problems in the legs or by a neurological disorder, although alcohol intoxication or malingering may occasionally cause difficulty. A full examination of the legs and feet should reveal any local cause, which may range from a painful corn to osteoarthritis of the hip. Abnormalities caused by neurological disorders are described in Chapter 16.



Figure 15.53 Hyperextensibility of the digits in Ehlers-Danlos syndrome.

Hypermobility

There is a wide variation in the range of normal joint movement, associated with age, sex and race. Excessive laxity or hypermobility of the joints (Fig. 15.53) can be defined in about 10% of healthy subjects and is frequently familial. It is also a feature of two inherited connective tissue disorders, Marfan syndrome and Ehlers-Danlos syndrome. Repeated trauma, haemarthrosis or dislocation may produce permanent joint damage. Hypermobility is often generalized but may occur at a single joint and must be considered if considerable joint laxity is demonstrated on examination. Hypermobility is often overlooked and should be considered in patients with joint pain with a lack of inflammatory or degenerative features on examination.

Work-related musculoskeletal disorders

Musculoskeletal pain arising as a result of a patient's occupation is an increasingly recognized cause of disability and economic loss. For example, spinal pain may be ascribed to poor seating in sedentary occupations. Likewise, an occupational history of repetitive movements may be relevant to upper limb pain. Assessment by an occupational health physician and an occupational therapist may be needed to consider workplace alterations.

Box 15.14 Common biochemical tests for inflammatory rheumatic conditions

Acute-phase reactant tests that are used in the assessment of inflammatory disease activity and in the subsequent monitoring of the patient:

- Erythrocyte sedimentation rate (ESR). This is a useful screening test, although it has poor specificity, being affected by the levels of haemoglobin, globulins and fibrinogen. Higher mean values are seen in healthy, elderly individuals.
- C-reactive protein (CRP). This is a more specific indicator of inflammation and is a good marker of the acute-phase response. A high ESR with a normal CRP is a useful pointer towards autoimmune rheumatic diseases, especially systemic lupus erythematosus (SLE).
- Plasma viscosity. This is a more specific measure of the acute-phase response than ESR, but may not be as widely available.
- Anaemia and thrombocytosis. Anaemia of chronic disease and a high platelet count often occur in inflammatory disease, but are non-specific. Other abnormalities on blood count, such as neutropenia, thrombocytopenia and lymphopenia, are common in SLE.
- Serum complement. Low levels of serum complement reflect activation owing to immune complex deposition; this may be a marker of disease activity in autoimmune diseases, such as SLE. Hereditary complement deficiencies are also associated with SLE.
- Muscle enzymes. Elevated creatine kinase levels occur in most patients with inflammatory myopathy.

Investigations in rheumatic diseases

When a full history and examination have been completed, investigations should be considered to support the working diagnosis or to distinguish between different possible diagnoses. They can be defined broadly as:

- Tests in support of inflammatory disease
- Diagnostic tests, including biopsies and imaging investigations

Tests in support of inflammatory disease

The acute-phase reactant tests that are used in the assessment of inflammatory disease activity and in the subsequent monitoring of the patient are listed in Box 15.14.

Diagnostic tests

Diagnostic tests differentiate between specific diseases and are relatively specific investigations.

Tests for rheumatoid factor

Rheumatoid factors are autoantibodies in the form of immunoglobulin (Ig) directed against other immunoglobulin G (IgG) molecules. IgM rheumatoid factor can be detected by its ability to clump particles coated with human IgG (latex test). This test is positive in about 80% of patients with rheumatoid arthritis. Results are reported as a titre, 1:80 or higher being a positive result. The original Rose-Waaler haemagglutination test used sheep erythrocytes coated with rabbit IgG to detect IgM (titres of 1:32 or more are positive); it has now been replaced by other tests. Enzyme-linked immunosorbent serum assay (ELISA) techniques are much more sensitive but produce positive results in many other conditions. Anti-citrullinated protein/peptide antibodies (ACPA) are a specific marker for rheumatoid arthritis and their presence indicates a poor prognosis, with patients at greater risk of joint damage, disability and loss of function.

These rheumatoid factor screening tests are useful where a diagnosis of rheumatoid arthritis is suspected, but they are not specific. Rheumatoid factor is frequently found in patients with other connective tissue diseases, for example SLE and Sjögren's syndrome, or other inflammatory disorders, such as subacute bacterial endocarditis and some viral infections.

Approximately 50% of patients with rheumatoid arthritis who present to a rheumatologist are positive for rheumatoid factor or anti-cyclic citrullinated peptide (CCP) antibodies when they initially present, rising to 70% to 80% at a later time point. Therefore, a negative serology test does not exclude rheumatoid arthritis as a diagnosis.

Antinuclear antibody tests

Antinuclear antibody (ANA), often referred to as antinuclear factor (ANF), is a very useful screening test for SLE as it is positive in up to 95% of patients. It is, however, non-specific, being positive in many other autoimmune rheumatic disorders, including about 20% of patients with rheumatoid arthritis. A positive test in children with arthritis may be associated with chronic iridocyclitis, which is frequently asymptomatic. Slit-lamp examination of the eye is mandatory to confirm the diagnosis.

The ANA test detects antibodies to intracellular nuclear and cytoplasmic antigens and is carried out by incubating the patient's serum with cells such as Hep-2 cells. After washing, a fluorescent antiserum to human IgG is used to detect human antibody adhering to the intracellular and nuclear antigens. A titre of 1:80 or more is significant, and adequate standardization is important.

DNA-binding test

Different immunochemical techniques (Farr assay or ELISA) may be used to detect antibodies to native double-stranded DNA. Another method is indirect immunofluorescence using the protozoon *Crithidia luciliae*, where the kinetoplast at the tail containing DNA fluoresces. The test is usually reserved for patients with a positive ANA test and is specific for SLE but not as sensitive as ANA. Occasionally it is positive in patients in whom the clinical suspicion of SLE is very high, but the ANA test is negative.

Box 15.15 Extractable nuclear antigen (ENA) tests

The following tests may be indicated by a speckled staining pattern in the antinuclear antibody (ANA) test. They can be summarized in terms of their clinical associations:

- Anti-Ro (SSA) and anti-La (SSB), typically in Sjögren's syndrome. Also seen in systemic lupus erythematosus (SLE), where they are associated with photosensitivity, and the neonatal lupus syndrome, which may result in congenital heart block and neonatal rashes.
- Anti-Sm in 5% to 10% of patients with SLE; a very specific marker if present.
- Anti-RNP (ribonucleoprotein) in some cases of SLE. Also picks out certain patients who have clinical features of other autoimmune rheumatic disorders and who are therefore often diagnosed as mixed connective-tissue disease (MCTD). The test can be considered a marker for the combination of clinical features, but the major clinical component of the condition will define management.
- Anticentromere antibody is found in limited cutaneous systemic sclerosis or 'CREST syndrome' (calcinosis, Raynaud's phenomenon, oesophageal symptoms, sclerodactyly and telangiectasiae).
- Anti-Scl 70 (DNA topoisomerase I) and anti-RNA polymerase are found in scleroderma and are associated with severe disease.

Box 15.16 Other helpful immunology tests in rheumatic disease

- Antineutrophil cytoplasmic antibodies are a marker for vasculitic conditions. Two immunofluorescence staining patterns occur:
  1. Cytoplasmic or c-ANCA, with specificity for proteinase 3 (a neutrophil enzyme), is specific for Wegener's granulomatosis (see Fig. 15.54)
- 2. Perinuclear or p-ANCA, with specificity for myeloperoxidase (and some other neutrophil enzymes), occurs in microscopic polyangiitis and other vasculitic diseases (and inflammatory bowel disease).
- Antiphospholipid antibodies, which include anticardiolipin antibodies, β2 glycoprotein 1 antibodies and the lupus anticoagulant tests, are associated with the antiphospholipid syndrome, characterized by arterial and venous thromboses and, in women, recurrent pregnancy loss and pregnancy morbidity. A false-positive VDRL test (see Chapter 18) may also be found in these patients.
- Anti-Jo 1 (histidyl t-RNA synthetase) is a marker for idiopathic inflammatory myopathies, such as dermatomyositis and polymyositis, especially when complicated by interstitial lung disease.
- Cryoglobulins are detected by clotting whole blood at 37°C and cooling the serum to 4°C and looking for a precipitate which is usually an IgM rheumatoid factor. Cryoglobulins are associated with vasculitis, infections such as hepatitis C and myeloproliferative disorders.
- Human leukocyte antigen (HLA) typing: the association of tissue antigen HLA-B27 with ankylosing spondylitis remains the strongest association in medicine. Although about 95% of patients with ankylosing spondylitis in the UK possess the B27 antigen, it is also found in 8% of the normal population. Ankylosing spondylitis, therefore, remains a clinical diagnosis, supported by typical radiographic findings. However, in early disease, in children with peripheral arthritis or where the clinical findings are atypical, HLA-B27 typing may provide supportive diagnostic value.
- Antistreptolysin-O (ASO) test: the presence in the serum of this antibody in a titre greater than 1/200, rising on repeat testing after about 2 weeks, indicates a recent haemolytic streptococcal infection.
- Viral titres. Certain viruses, notably parvovirus and Coxsackie virus, may cause transient musculoskeletal symptoms that may be mistaken for systemic diseases. Rising viral titres may be useful in the differential diagnosis.

Antibody tests to extractable nuclear antigens (ENA)

Box 15.15 shows tests that may be indicated by a speckled staining pattern in the ANA test. They can be summarized in terms of their clinical associations. Other diagnostic antibody tests are shown in Box 15.16.

Uric acid

A consistently normal plasma uric acid level (<375 mmol/l in women, <425 mmol/l in men) usually excludes the diagnosis of untreated gout. Raised levels occur in many circumstances and do not in themselves establish the diagnosis of gout (see later). On a low-purine diet, the 24-hour urinary urate excretion should not exceed 600 mg. Higher levels indicate 'overproduction' of urate and a risk of renal stone formation.

Synovial fluid examination

Synovial fluid may be obtained for examination from any joint in which it is clinically detectable. The knee is the most convenient source: after infiltration with a local anaesthetic, a 21-gauge needle is inserted into the joint between the patella and the femoral condyle. For smaller, more inaccessible joints, such as the wrist or MCP, an ultrasound-guided aspiration should be considered. The aspirated fluid should be placed in a plain sterile container; if a cell count is required, some of the fluid should be mixed with ethylenediaminetetra-acetic acid (EDTA) anticoagulant.

An injured joint can also be aspirated (Box 15.17). The joint swollen after injury may reveal clear pink fluid, suggesting a meniscal lesion, or show frank blood. The latter is usually indicative of a torn anterior cruciate ligament. If blood is aspirated, look at its surface for fat globules. This is derived from the marrow and confirms an intra-articular fracture. Synovial fluid examination is diagnostic in two conditions—bacterial infections and crystal synovitis—and every effort should be made to obtain fluid when either of these is suspected. Polarized light microscopy can differentiate between the crystals of urate in gout and those of calcium pyrophosphate dihydrate in pseudogout. Outside these conditions, synovial fluid examination is unlikely to be diagnostic. Frank blood may point to trauma, haemophilia or villonodular synovitis,

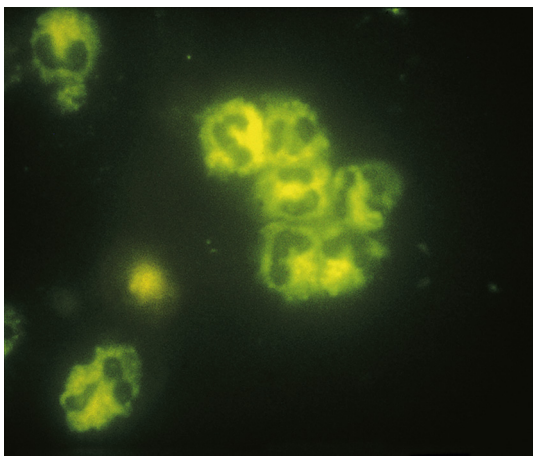


Figure 15.54 Antineutrophil cytoplasmic autoantibodies with cytoplasmic staining (c-ANCA). This pattern has a high predictive value for a diagnosis of Wegener's granulomatosis.

Box 15.17 Findings in synovial fluid after aspiration of the knee joint

- Cloudy fluid or pus: bacterial infection (see text)
- Urate or pyrophosphate crystals: gout or pseudogout
- Pink fluid: torn meniscus
- Blood: trauma, haemophilia, villonodular synovitis

Box 15.18 Biopsy procedures and tissue harvesting that may be useful in the differential diagnosis of rheumatic diseases

- Muscle biopsy should be considered in patients with inflammatory myopathies, such as dermatomyositis, polymyositis and inclusion body myositis. Electromyography may show typical features of myopathy and biopsy may be guided by magnetic resonance imaging (MRI) of muscles (see Fig. 15.55).
- Rectal biopsy can be useful in the diagnosis of amyloidosis secondary to chronic inflammatory disease, but renal biopsy may still be necessary if the cause of renal impairment is not clear.
- Renal biopsy is essential in the vasculitides or systemic lupus erythematosus (SLE) where active glomerulonephritis is suspected. Vasculitis may also be confirmed on renal biopsy but, in general, tissues found to be abnormal on clinical examination or by further investigation (e.g. skin, muscle, sural nerve or liver) should be considered first for diagnostic biopsy in undifferentiated systemic vasculitis.
- Biopsy of a lip minor salivary gland may be useful to confirm Sjögren's syndrome.
- Temporal artery biopsy is often diagnostic in patients with clinical features of temporal (giant cell) arteritis. This is the investigation of choice.
- Synovial biopsy is of little value in the differential diagnosis of inflammatory polyarthritis, but should be considered in any unusual monoarthritis to exclude infection, particularly tuberculous, or rare conditions such as sarcoid, amyloid arthropathy or villonodular synovitis. This may be performed by arthroscopy or using a semi-automated guillotine needle under ultrasound guidance.
- Bone biopsy. May be useful in the diagnosis of osteomalacia, malignancy, renal osteodystrophy and Paget's disease. The bone marrow can also be aspirated at the same time if necessary.

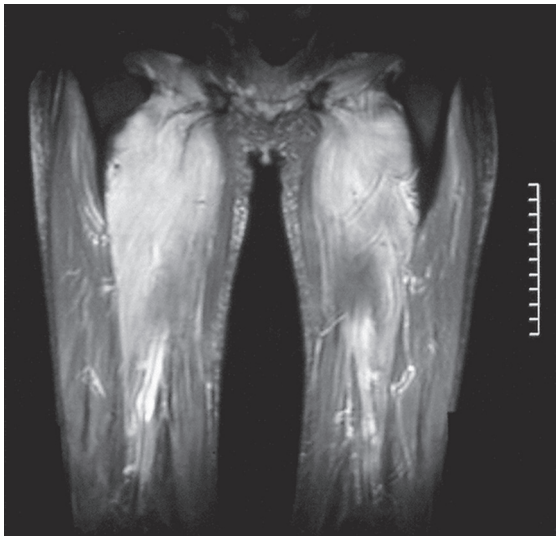


Figure 15.55 Magnetic resonance imaging (MRI) scan of thigh muscles showing high signal lesions of inflammatory myositis.

Box 15.19 General principles of musculoskeletal imaging

- Use a systematic approach (see Box 15.13).
- Age, sex and clinical information are essential in interpretation.
- Radiographs reveal bones and soft tissues.
- Always obtain two views, at right angles, in trauma patients.
- Radiographs may be normal even in the presence of disease.
- Diffuse abnormalities are difficult to detect.
- Bone-based and joint-based disease must be differentiated.
- Normal variants can be confused with pathology.

whereas inflammatory (as opposed to degenerative) arthritis is suggested by opaque fluid of low viscosity, with a total white cell count greater than 1000/ml, neutrophils more than 50%, protein content more than 35 g/l and the presence of a firm clot. Culture of this fluid may produce a bacterial growth, usually of staphylococci, but occasionally *Mycobacterium tuberculosis* or other organisms.

Biopsies useful in differential diagnosis

Box 15.18 shows tests that may be useful in the differential diagnosis of rheumatic diseases.

Radiological examination

Certain general principles are important, particularly as clinicians may be asked to give an opinion on

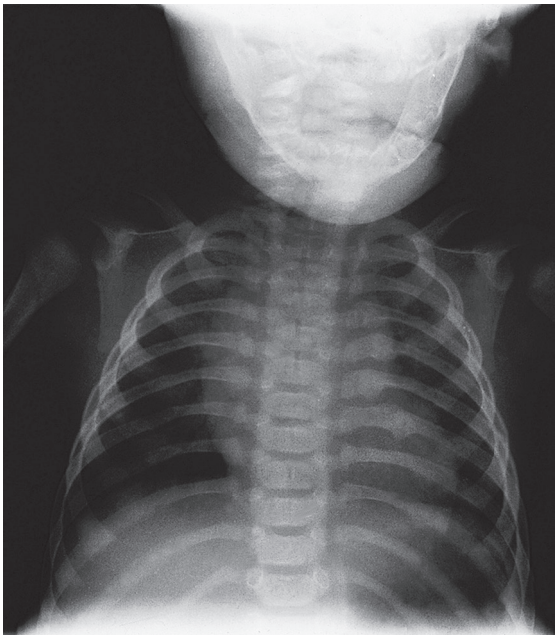


Figure 15.56 Healed fracture of posterior left and right ribs in a 6-month-old infant, classic non-accidental injury (NAI).

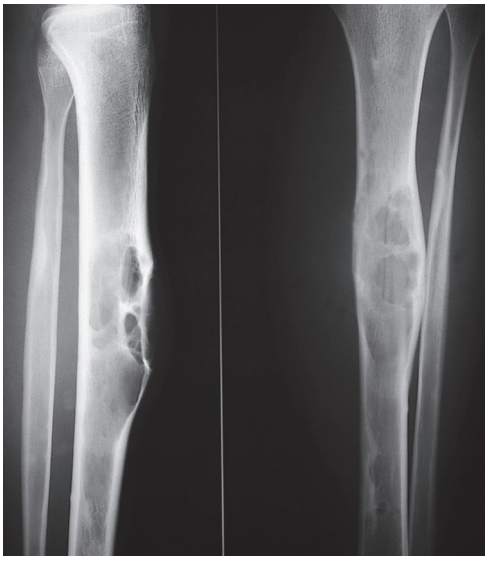


Figure 15.57 Benign bone tumour (adamantinoma). Note the solitary well-defined lucent lesion with a sclerotic margin and the absence of matrix calcification.

radiographs of limbs after a traumatic injury, whether minor or more serious (Box 15.19). Common problems are shown in Figures 15.56–15.59. A systematic approach is essential (Box 15.20).

Bone density (Table 15.6) may be normal, reduced (osteopenia) or increased (osteosclerosis). These changes are easy to detect if focal, but difficult if diffuse. When a focal bone lesion (Box 15.21) is noted, look at its position in the bone and at its margins, and note whether there is any focal matrix

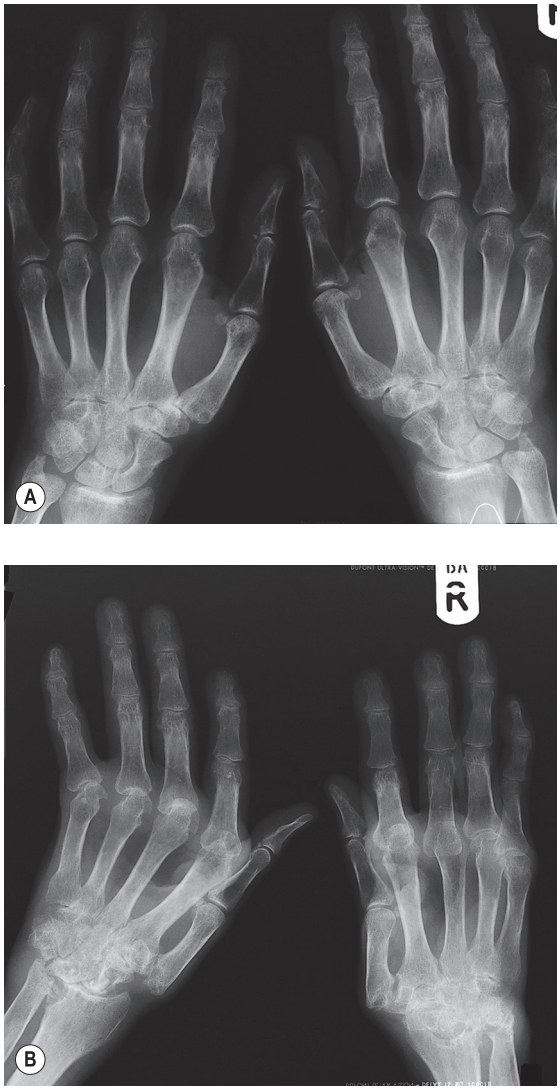


Figure 15.58 **(A)** Early rheumatoid arthritis: local osteopenia, loss of joint spaces, soft-tissue swelling. **(B)** Late rheumatoid arthritis: erosion of periarticular surfaces and ulnar subluxation.



Figure 15.59 Renal osteodystrophy: generalized demineralization, terminal phalangeal and subperiosteal bone resorption and vascular calcification.

Box 15.20 Systematic approach to musculoskeletal imaging

- Bone density
- Soft tissues
- Joints
- Bone
- Periosteum

calcification, whether the cortex of the bone is intact and whether there is any periosteal reaction around it. Most solitary bone lesions in young people are benign and show no periosteal reaction or swelling around them, and no associated soft-tissue swelling. Aggressive (malignant) bone lesions are more common in the elderly. Certain bone metastases have a characteristic appearance (Box 15.22). Isotope imaging is useful in detecting multiple sites of bony involvement in generalized disease and in metastatic cancer. Computed tomography (CT) imaging is also sensitive in detecting and analysing bony lesions, because it provides good images of the bony margins of the lesions and of the associated soft tissues. MRI is used to assess the extent of the lesion and any local soft-tissue invasion.

The most common use of plain radiographs is to document joint pathology, such as osteoarthritis (joint space narrowing, periarticular sclerosis, subchondral cysts and osteophytes), or inflammatory conditions, such as rheumatoid arthritis (joint space narrowing, periarticular osteopenia, erosions and subsequent bony ankylosis). Imaging of the hands and feet is most commonly performed when investigating an inflammatory arthritis, as subclinical inflammation and joint description may occur in the absence of frank joint inflammation. Osteoarthritis requires a more focused approach to imaging of symptomatic joints. Although imaging changes characteristic of degenerative joint disease can be detected by plain radiographs, they do not correlate well with symptom severity.

Fractures

X-rays are often the first investigation in suspected fractures and in joint disease. In traumatic fractures, X-rays are diagnostic and are used to check alignment of the fracture and healing. X-rays of a fracture are also important in excluding a pathological fracture associated with metabolic bone disease, a focal benign bone lesion or neoplastic invasion by metastases. When there is clinical doubt after

Table 15.6 Abnormal bone density

Generalized osteopenia	Generalized osteosclerosis	Benign focal lucent lesions	Benign focal sclerotic lesions
Ageing osteoporosis	Osteopetrosis (marble bone disease)	Simple bone cyst, aneurysmal bone cyst	Bone island
	Metastatic bone disease, e.g. prostate and breast cancer	Fibrous cortical defect	Callus after fracture
Disuse, e.g. trauma and neurogenic paralysis, osteogenesis imperfecta	Dietary causes, e.g. hypervitaminosis A, fluorosis	Non-ossifying fibroma	Paget's disease of bone
Acquired metabolic bone disease, e.g. rickets, osteomalacia, hyperparathyroidism	Acquired metabolic bone disease, e.g. renal osteodystrophy	Enchondroma	Bone infarction Osteoid osteoma
Myeloma	Myelofibrosis, sickle cell disease	Fibrous dysplasia, giant cell tumour of bone	Fibrous dysplasia

Box 15.21 Benign and malignant bone lesions

Benign

- Young person
- Single lesion
- Well-defined margin
- Intact cortex
- No periosteal reaction
- No growth
- Asymptomatic

Malignant

- Older age
- Multiple lesions
- Poorly defined margin
- Destroyed cortex
- Periosteal reaction
- Soft-tissue extension of lesion
- Lesions painful

Box 15.22 Bone metastases

- Expansile: thyroid, kidney, breast, bronchus, melanoma and myeloma
- Sclerotic: prostate and breast
- Lytic: breast, bronchus, kidney, thyroid and melanoma
- Mixed: breast, bladder or previously treated (irradiated) bone lesions

a non-diagnostic X-ray, an MRI scan often reveals the underlying fracture and is the investigation of choice. Radiographs taken to confirm or exclude a fracture must be taken in two planes. It is essential that either the whole limb is turned or the imaging equipment rotated. The limb must not be twisted at the fracture site. When looking for a fracture, run a pen tip or its equivalent around the cortex of the bone as seen on the film (without leaving marks). Any break in continuity will reveal itself; do not confuse an epiphysis with a fracture. Note soft-tissue swelling and distension of joints. Always seek a radiology opinion if in doubt.

In joint disease, MRI is the imaging method of choice because it can visualize all the soft-tissue components of the joints. Osteopenia is a non-specific feature of disuse, but it also occurs in relation to affected joints in rheumatoid arthritis and Still's disease of children. Involvement of the distal interphalangeal joint is a feature of psoriatic arthropathy. In rheumatoid disease, the involvement of joints is usually symmetrical, and the wrist is particularly susceptible.

Only radiographs likely to yield specific information should be requested. However, in unilateral joint disease, it is useful to examine both sides for comparison (Fig. 15.60). In patients with inflammatory polyarthritis, three routine films are helpful in the diagnosis and assessment of progression, with both hands and wrists on one plate and both feet on another to compare bone density and to look for periosteal reaction or erosive change, and one of the full pelvis (Fig. 15.61) to show the sacroiliac and hip joints. In the absence of so-called 'red flag' signs, for example weight loss, night pain, fevers or neurological signs, most spinal radiographs are unnecessary.

Specialized radiology

The following imaging techniques can provide precise information about localized pathology, but they are dependent on the clinician making a clear diagnostic request with as much clinical information as possible:

- High-resolution ultrasound is of value in defining soft-tissue structures, including muscles and tendons; it provides an excellent means for





Figure 15.60 Left and right knee joints. The X-rays show normal joint anatomy, but with chondrocalcinosis **(A)** and osteoarthritic change **(B)** on the opposite side. Note the increased bone density and narrowing of the joint space.

guiding aspiration of joint effusions and biopsy procedures. Ultrasound consistently has been demonstrated to be more sensitive than clinical

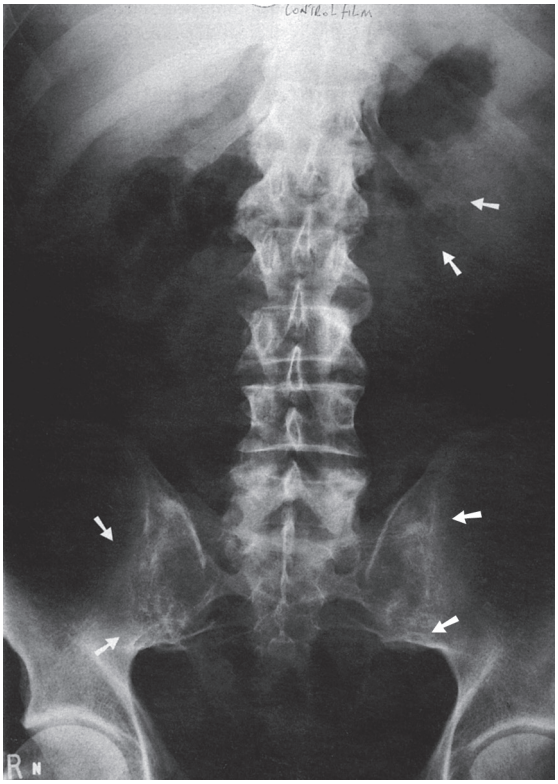


Figure 15.61 X-ray of lumbosacral spine and upper pelvis. There is fusion of several vertebrae, and of the sacroiliac joints *(arrows)* from ankylosing spondylitis. The renal papillae on the left are calcified, evidence of previous papillary necrosis from prolonged analgesic use.

examination in the context of joint inflammation and it provides considerable additional information in the diagnosis and management of inflammatory arthritis. Grey-scale synovial thickening and power Doppler signal reflect histologically defined synovitis and can be used to differentiate between true joint pathology and fibromyalgia and other pain syndromes (Fig. 15.62). In addition, this imaging modality has the unique capacity to be used in a dynamic fashion, allowing tendon, ligament and joint function to be observed in real time. Rotator cuff pathology at the shoulder is amenable to ultrasound imaging and is often the first imaging investigation of choice when assessing the integrity of the muscle and tendons of these structures. Ultrasound can detect erosions and joint damage at a very early stage of disease before X-ray changes are demonstrated.

- CT. The combination of superior tissue contrast and tomographic technique permits definition of soft-tissue structures obscured by overlapping structures, including intervertebral discs and other joints normally difficult to visualize, such as sacroiliac (Fig. 15.63), sternoclavicular and subtalar. Bone pathology is particularly well

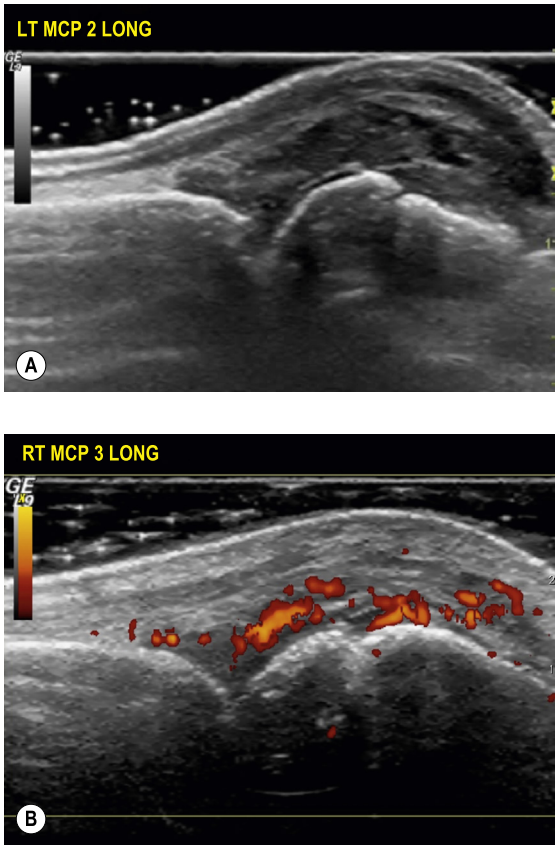


Figure 15.62 **(A)** Synovial thickening seen as hypoechoic *(darker grey)* material over the dorsal aspect of the metacarpal head and neck using ultrasound. **(B)** Power Doppler signal seen as *red/orange* signal located within the synovial thickening within the metacarpal-phalangeal joint.



Figure 15.63 Computed tomography (CT) scan of sacroiliac joints, showing a distinctive lesion on the left side with a sequestrum owing to tuberculosis.



Figure 15.64 A lumbar spine magnetic resonance imaging (MRI) scan. There is a disc protrusion at L4/5 with degeneration of the disc itself, shown by the less bright signal in the intervertebral disc at this level.

characterized on CT imaging (e.g. malignancy, erosive disease or fracture). MRI has unique advantages in evaluating the musculoskeletal system, but it is vital that the clinician defines the pathology suspected, as correct positioning and sequence selection (T-weighting) are vital in optimizing image quality. MRI is increasingly used to image the major joints in the limbs, especially the knee (see Fig. 15.47), hip, shoulder and elbow.

In the context of inflammatory arthritis, MRI has much to recommend it as an imaging modality. There is a clear relationship with bone marrow oedema on MRI and the subsequent development of erosive arthritis providing an early indicator of potentially poor prognosis. Erosive disease, when present, is detected earlier than with plain radiographs and changes in synovial inflammation can be readily detected after therapeutic intervention. A standardized, validated scoring system has been developed (RAMRIS: *R*heumatoid *A*rthritis *MRI* Scoring), which encompasses multiple elements of MRI-detected synovial inflammation and joint damage.

Enhancement by the use of intravenous paramagnetic contrast (e.g. gadolinium) has further improved definition in spinal imaging.

Magnetic resonance imaging is invaluable when assessing the degree of spinal involvement in spondyloarthropathies, such as ankylosing spondylitis, particularly in early disease where radiography ankylosis is absent. Both costovertebral

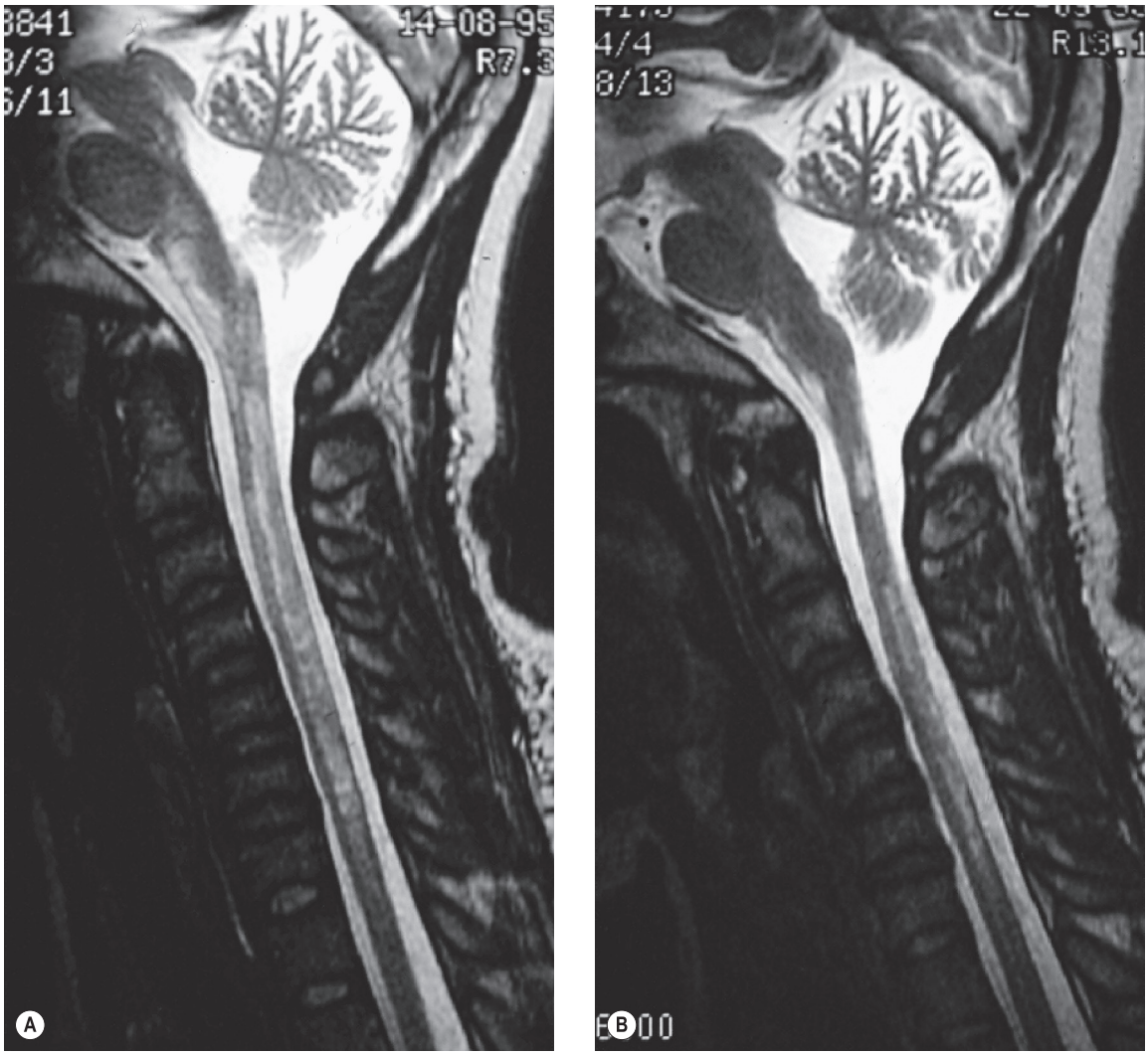


Figure 15.65 Magnetic resonance image (MRI) of cervical spine in a patient with transverse myelitis owing to systemic lupus erythematosus (SLE) **(A)** before and **(B)** after immunosuppression.

and sacroiliac joints are readily imaged with good sensitivity for inflammatory pathology.

It is of particular value in the non-invasive investigation of disc disease (Fig. 15.64), including spinal infection, and is a sensitive technique for the diagnosis of avascular necrosis. MRI scanning of the brain and spine is the single most important investigation in neuroimaging, as the neural tissues themselves, together with supporting tissues, are well visualized. Bone is less well seen with this technique. As a rule of thumb, most abnormalities appear dark on T1-weighted images and white on T2-weighted images. Some common problems are shown in Figures 15.55, 15.64 and 15.65.

- Isotopic scanning (scintigraphy) can be used in the diagnosis of acute (e.g. infection or stress fracture) or multiple (e.g. metastases) bone lesions by use of the first 2-minute (dynamic blood flow) phase, second 10-minute (blood pool) phase and third 3-hour late phase (osteoblastic) (Fig. 15.66) following intravenous injection of diphosphonate compounds. Tomographic scintigraphy can further refine definition of the isotope uptake (e.g. in stress fracture of the pars interarticularis). Radionucleotide bone scans are more sensitive than plain radiographs for the diagnosis of Paget's disease and in polyostotic cases help to quantify the overall burden of disease. Scintigraphy has a high sensitivity to metabolically active lesions; however, it is relatively non-specific and is often used in conjunction with other imaging modalities.
- Dual-energy X-ray absorptiometry (DEXA) is widely used to assess bone mineral density in metabolic bone disease and osteoporosis. Scans recorded with long intervals (e.g. every 2 to 3 years) between them may be used to assess the

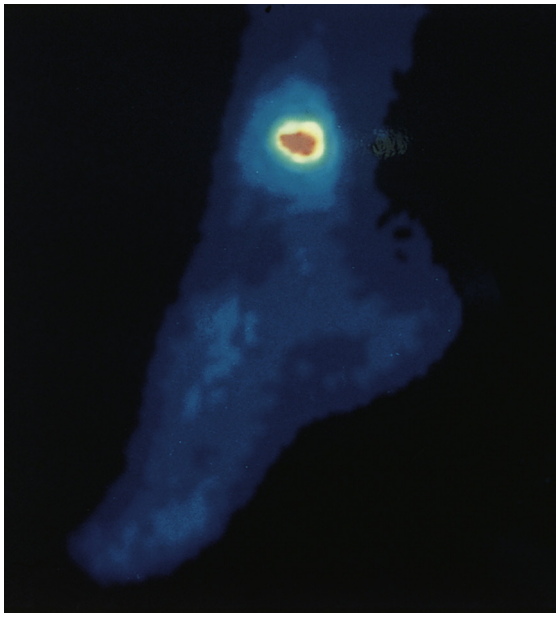


Figure 15.66 Technetium bone scan in a distance runner, showing focally increased uptake in the lower tibia owing to a stress fracture.

effects of therapy. Results are expressed in relation to a normal reference range.

- Positron emission tomography (PET) scanning is becoming more widely available, but remains expensive. This is a type of scintigraphic molecular imaging based mostly on the uptake of radiolabelled glucose by metabolically active structures. PET scans are mostly used in rheumatology investigations and have better signal-to-noise ratio than standard isotope bone scanning. The combined technology of CT and PET scanning allows the acquisition of structural information simultaneously with metabolic activity. It is extremely sensitive in detecting malignancy and is also proving useful in documenting large vessel inflammation in disorders such as Takayasu's arteritis and giant cell arteritis. The radiation exposure is considerable and, as such, this imaging modality should be used cautiously and appropriately.